

One Geometry, Two Research Questions

Superposition, robustness
and recursive-training stability

A derivation companion to the monosemanticity paper

Exact toy results, proofs, counterexamples, numerical checks,
and a route to testing the same mechanism in diffusion models.

Project 1: the superposition-to-robustness phase diagram.

Project 2: model collapse as monosemanticity decay.

Structure: mathematics $\rightarrow$ controlled toy models $\rightarrow$ real-model validation.

Horizon: eight-week research core, with a two-week extension if available.

Prepared: 20 September 2026; Project 1 derivations updated 6 October 2026. Conditional derivations are distinguished from empirical hypotheses and from claims of research novelty.

What this volume delivers

This volume keeps the two agreed projects and their shared monosemanticity motivation. It does not replace them with a benchmark, an agent system or a different application area. It supplies a mathematically explicit foundation that can support their development.

The exact formulas delivered here include: operational perturbation radii; reconstruction risk; a finite-state feature-detection risk; a small, nonlinear superposition phase boundary with feature frequency and importance; a general interference-based error bound; exact recursive toy dynamics; a bound connecting one-generation feature errors to cumulative distribution drift; a local derivative for a genuinely retrained model; and a Gaussian denoising calculation clarifying the diffusion connection.

These are *finished conditional calculations*, not a claim that the proposed conference contribution is already established. Some are standard facts presented with proofs; others are worked combinations specialized to this program. Their novelty has not been certified. The general learned-geometry phase diagram and a validated monosemanticity-based predictor in real diffusion models remain the research targets.

That distinction is essential to using these notes productively. A theorem can be correct and still be too elementary for a new paper. A promising experiment can still contradict a preferred mechanism. The purpose of the derivations is to make the remaining research time test precise claims, not to promise an outcome that has not been observed.

Project 1 update: exact scope of the completed mathematics

The added clean-training and noise chapters derive the finite bias profiles, dense-case optima, a certified low-frequency geometry, the global clean frequency transition, and the at-least-two noise-crossing result. They preserve the distinction between reconstruction-selected representations and downstream binary detection. All new mathematics remains in this volume rather than only in working Markdown files. The arbitrary-load/importance target and real-model transfer are not complete; the original Project 2 material is preserved as background, with no new recursive-training experiment introduced.

The scientific question in plain language

When a model shares representational directions between concepts, it can store more information in limited space. That sharing can also create confusion. We want to understand the conditions under which the gain in coverage outweighs the cost in errors, and whether the same errors accumulate when generated outputs become future training data.

This is a general learning-and-representation question. Safety is a possible application, but no adversary is needed for the main study. Recursive training supplies the connection to the team’s self-improvement interests; a self-consuming loop does not count as improvement unless quality actually improves. Vision diffusion provides a real generative setting in which to test the mechanism.

Read these chapters in sequence

Chapters 1–3 establish definitions and operational interpretations. Chapters 4–6 derive fixed-geometry Project 1 results and a general interference bound. Chapters 7–8 derive clean-selected geometry, the global frequency transition and the certified nonmonotone noise comparison. Chapters 9–12 derive Project 2 and the mathematical connection. Chapters 13–15 explain diffusion, controlled validation and the original eight-week execution outline. The appendices give a formula ledger, exercises, numerical results and primary references.

No team member is assigned ownership in this document. The work has a continuous theoretical and experimental thread, with room for complementary interests in interpretation, robustness, evaluation, post-training and generative modeling.

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Chapter 1

Fix the question before manipulating the symbols

Learning objective. Specify the prediction task, corruption, representation and training loop so a formula has a definite meaning.

1.1 The commitments that remain fixed

Project 1 seeks a robust-performance tradeoff in sparsity, feature-to-dimension load and importance decay. Project 2 seeks a representation-level account of recursive-training degradation and a quantity that predicts it. Their intended combination uses shared feature geometry, shared toy infrastructure and a real-model demonstration. The source anchor is Zhang et al.’s monosemanticity-and-robustness paper, taught in Volume I.

The mathematical choices below instantiate this direction; they do not exhaust it. We begin with a task whose ground truth is unambiguous: recovering binary feature presence. This allows an actual error probability to replace an informal “cleanness” score. Continuous-amplitude reconstruction is also derived as a companion task.

1.2 The variables

Let $b = (b_1, \dots, b_n) \in \{0, 1\}^n$ denote genuine toy concepts. Initially $b_i \sim \text{Bernoulli}(p_i)$ independently, unless otherwise stated. Let $s_i = 1 - p_i$ and let $I_i > 0$ weight errors. The encoder is $W \in \mathbb{R}^{m \times n}$, with columns w_i . Define

$$h = Wb, \quad G = W^\top W. \quad (1.1)$$

The load is n/m . Later, a training generation is indexed by g , and the geometry may become W_g, G_g . A diffusion denoising step uses a different index τ .

1.3 Three noise locations

1. **Feature-space corruption:** b or a continuous feature vector is perturbed before encoding. A feature-space adversary is an idealized mathematical model.
2. **Code-space corruption:** $h = Wb + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I_m)$. This makes each matched readout’s variance depend on G_{ii} .
3. **Observed-input corruption:** pixels or waveforms change before a nonlinear encoder. The encoder’s Jacobian and nonlinear activation changes matter.

These yield different formulas. A proof for the first cannot be described as a guarantee for the third without an additional argument.

1.4 Three outcomes

Feature detection error asks whether a concept is correctly declared present. **Reconstruction error** asks how accurately its amplitude is recovered. **Distributional fidelity** asks whether a generator produces the correct frequencies and combinations of concepts. A classifier can retain accuracy while a generator loses diversity, so these outcomes should be linked mathematically rather than treated as interchangeable.

For binary detection, the main risk is

$$R(W, \theta; P) = \sum_i I_i \mathbb{P}_{b \sim P, \xi} \{\hat{b}_i \neq b_i\}. \quad (1.2)$$

For a phase diagram, compare the risks of two admissible representation families at the same stated resource budget. A boundary is the set where their risks are equal. Calling it a “phase boundary” is descriptive here; a finite toy model does not automatically have a thermodynamic singularity.

Before moving on. For every proposed formula, write one sentence specifying what is random, what is fixed and what outcome it predicts.

What the original margin formula really means

Learning objective. Retain the valid geometric identity while making its adversarial interpretation exact.

2.1 The general linear decision radius

Consider a binary decision with label $y \in \{-1, 1\}$ and score $f(x) = \beta^\top x - c$, where $\beta \neq 0$. If $yf(x) > 0$, the example is correctly classified. For a perturbation with $\|\delta\|_q \leq \varepsilon$, Hölder's inequality gives

$$\min_{\|\delta\|_q \leq \varepsilon} yf(x + \delta) = yf(x) - \varepsilon \|\beta\|_{q^*}, \quad \frac{1}{q} + \frac{1}{q^*} = 1. \quad (2.1)$$

Equality is attained by a dual-norm maximizing direction. Therefore the exact distance to the decision boundary is

$$r_q(x, y) = \frac{yf(x)}{\|\beta\|_{q^*}}. \quad (2.2)$$

This assumes unrestricted perturbations in the chosen vector space. If valid inputs must remain nonnegative, binary or on a manifold, the feasible attack set changes and this may become a lower bound on the constrained radius rather than the exact answer.

2.2 A normalized matched-readout example

For the tied linear readout of feature i , take $\beta = Ge_i$, $x = e_i$, $c = 0$, $y = 1$ and $q = 2$. Then

$$M_i = \frac{G_{ii}}{\sqrt{\sum_j G_{ij}^2}} = \left(1 + \sum_{j \neq i} \frac{G_{ij}^2}{G_{ii}^2}\right)^{-1/2}. \quad (2.3)$$

This requires $G_{ii} > 0$. If the feature is dropped, the entire row is zero and the ratio is 0/0, not a meaningful robustness score.

Equation (2.3) is also the cosine between the row of G and the target feature axis. It is a legitimate geometric quantity. It equals an attack radius only in the normalized input and decision problem just specified. A different input x , threshold c or feature amplitude changes the radius through Equation (2.2).

The squared alignment score also equals the established per-feature capacity in Scherlis et al., *Polysemanticity and Capacity in Neural Networks*, Equation (3):

$$C_i = \frac{(w_i^\top w_i)^2}{\sum_j (w_i^\top w_j)^2} = M_i^2. \quad (2.4)$$

This identity prevents treating the score itself as a new contribution. A dropped feature requires a separate convention because the ratio is undefined.

2.3 It is a matched readout, not always the optimal readout

The choice $w_i^\top h$ corresponds to the tied decoder. A general linear estimator of x_i from h need not use w_i . For centered data with covariance C and additive code-noise covariance N , the least-squares coefficient is

$$a_i^* = (WCW^\top + N)^\dagger WCe_i, \quad (2.5)$$

with the appropriate identifiability conditions or pseudoinverse convention. Its effective feature-space direction is $W^\top a_i^*$. Calling Ge_i “the optimal readout” without these qualifications was unjustified.

2.4 Interpreting isolation, rarity and signed overlap

Can an attacker move a perfectly isolated feature? Yes. If $G = I$, then $M_i = 1$, but changing x_i changes its reading. An ability to move the score does not imply $M_i < 1$.

Does a negative overlap count? In this unrestricted worst-case ℓ_2 setting, G_{ij} enters squared, because an adversary can choose the harmful sign. For ordinary nonnegative data and a threshold decoder, the sign can be beneficial or harmful depending on co-occurrence. The square loses that distinction.

Does rarity determine M_i ? No. At fixed G , Equation (2.3) has no activation probability in it. Rarity may affect what G learns, or statistical collision probabilities, but it cannot be inserted into this fixed-geometry identity by multiplication.

Is M_i scale invariant? It is invariant under a common rescaling $W \mapsto aW$, since all entries of G scale by a^2 . Rescaling only one column changes its overlaps relative to its diagonal and generally changes M_i . The diagonal has not disappeared physically merely because it is hidden in normalized overlaps.

2.5 Other threat models

For a code-space perturbation $h \mapsto h + \delta_h$, the tied readout changes by $w_i^\top \delta_h$, so its worst-case swing is $\varepsilon \|w_i\|_2$. For random feature noise with covariance N_x , the output variance is $(Ge_i)^\top N_x (Ge_i)$. These differ from each other and from the feature-space worst-case swing.

For a nonlinear observed-input map $h = f(u)$, a local first-order calculation uses $\nabla_u[w_i^\top f(u)]$. To obtain a finite-radius guarantee, bound the remainder or a Jacobian norm over the entire perturbation region. A Jacobian at one point is a local sensitivity measurement, not automatically a certificate.

Status of this result. Equations in this chapter are standard linear estimation and norm-duality facts. They repair the operational meaning of the earlier margin; they are not, by themselves, a novel conference contribution.

Chapter 3

Linear reconstruction: the useful calculation and its limit

Learning objective. Put sparsity and importance into a risk correctly, and see why a purely linear model can miss the desired tradeoff.

3.1 The exact reconstruction risk

Let x have mean μ and covariance C . Observe $x + \epsilon$, where ϵ has mean zero, covariance N and is independent of x . Reconstruct $\hat{x} = G(x + \epsilon)$. Set $D = \text{diag}(I_1, \dots, I_n)$. Then

$$L = \mathbb{E}[(x - \hat{x})^\top D(x - \hat{x})] \quad (3.1)$$

$$= \text{tr}[D\{(I - G)C(I - G)^\top + GNG^\top\}] + \|D^{1/2}(I - G)\mu\|_2^2. \quad (3.2)$$

To prove this, write the error as $(I - G)(x - \mu) + (I - G)\mu - G\epsilon$. Cross terms involving centered independent variables vanish. Use $\mathbb{E}[z^\top Dz] = \text{tr}(D \text{Cov } z) + (\mathbb{E}z)^\top D(\mathbb{E}z)$.

For independent spike-and-uniform features, $\mu_i = p_i/2$ and $C_{ii} = p_i/3 - p_i^2/4$. For binary features, $\mu_i = p_i$ and $C_{ii} = p_i(1 - p_i)$. Importance weights multiply the error after these distributional quantities are specified. This is an exact place for rarity and importance to enter, unlike inserting p into Equation (2.3).

3.2 An isotropic example with no distributional advantage

Take centered, isotropic features $C = vI_n$, isotropic input noise $N = \sigma_x^2 I_n$ and equal importance. Let F be the Gram matrix of a unit-column tight frame, so

$$\text{diag } F = \mathbf{1}, \quad F^2 = \frac{n}{m}F, \quad \text{rank } F = m. \quad (3.3)$$

Use $G = gF$. Every feature sees aggregate squared overlap $E = n/m - 1$, and its reconstruction risk is

$$\ell(g) = v(1 - g)^2 + vg^2E + \sigma_x^2g^2(1 + E). \quad (3.4)$$

Differentiate, set the derivative to zero, and obtain

$$g_* = \frac{v}{(v + \sigma_x^2)(1 + E)}. \quad (3.5)$$

The minimized total risk is

$$L_* = nv - \frac{mv^2}{v + \sigma_x^2}. \quad (3.6)$$

Now compare a monosemantic linear representation retaining m coordinates, each with optimal shrinkage $a = v/(v + \sigma_x^2)$, and dropping the rest. Each retained coordinate has risk $v\sigma_x^2/(v + \sigma_x^2)$; each dropped coordinate costs v . Their sum is the same expression.

This is not evidence against the project. It shows that isotropic second moments with a linear decoder can erase the sparse, nonlinear mechanism of interest. The learned sparse-superposition phenomenon cannot be assumed to survive replacing its decoder by an arbitrary linear map.

3.3 A controlled way forward

Keep a threshold or ReLU decoder, retain ground-truth concepts, and include the risk of dropped features in the comparator. The next chapters do this. The linear identities remain useful as sanity checks and baselines, but the nonlinear feature-detection model produces an actual sparsity-dependent tradeoff.

Before moving on. Explain why two distributions with identical mean and covariance have identical risk in Equation (3.2), but can behave differently after a nonlinear decoder.

Project 1: an exact risk for arbitrary fixed geometry

Learning objective. Derive an operational error probability that contains feature frequency, importance and geometry explicitly.

4.1 A matched feature detector

Observe $h = Wb + \xi$ with $\xi \sim \mathcal{N}(0, \sigma^2 I_m)$. Read feature i with

$$z_i = w_i^\top h = (Gb)_i + \zeta_i, \quad \hat{b}_i = \mathbf{1}\{z_i > \theta_i\}. \quad (4.1)$$

The vector $\zeta = W^\top \xi$ is Gaussian with covariance $\sigma^2 G$. Its components are generally correlated. For a single detector, its marginal variance is $\sigma^2 G_{ii}$.

Let Φ denote the standard normal CDF. Conditional on a fixed binary state b , the error probability is

$$e_i(b) = \Phi\left(-\frac{(2b_i - 1)((Gb)_i - \theta_i)}{\sigma\sqrt{G_{ii}}}\right). \quad (4.2)$$

When $b_i = 1$, failure means the Gaussian score lies below threshold. When $b_i = 0$, failure means it lies above threshold. These are exactly the two signs in the formula.

Proposition 4.1 (Exact finite-state weighted risk). *For $\sigma > 0$ and all scored columns satisfying $G_{ii} > 0$,*

$$R(G, \theta; P) = \sum_{i=1}^n I_i \sum_{b \in \{0,1\}^n} P(b) e_i(b). \quad (4.3)$$

For an initially independent distribution, $P(b) = \prod_j p_j^{b_j} (1 - p_j)^{1-b_j}$.

Proof. Condition each indicator of error on b , use Equation (4.2), and apply linearity of expectation. No independence between detector errors is needed. Independence of input features is needed only for the product formula for P . $\square$

4.2 What the formula buys us

This is an exact model-based prediction, not a regression fit. For small n , enumerate states and calculate risk directly. For larger n , Monte Carlo estimates the same defined quantity. Importance decay enters as $I_i = i^{-\gamma}$, frequency enters through P and capacity enters through the admissible rank- m geometry.

A dropped feature requires separate treatment. A detector constantly predicting zero has error p_i ; a best constant detector has error $\min(p_i, 1 - p_i)$. The choice must be fixed when defining the comparison. We use the zero-output reconstruction convention below, appropriate to representing absence when a concept is omitted.

4.3 Rarity also affects the best decision threshold

The midpoint thresholds used in the explicit pair calculation are fixed detector choices, not a claim of Bayes optimality for every prior. For an isolated scalar code $h = ab + \xi$ with $b \sim \text{Bernoulli}(p)$, $0 < p < 1$, and Gaussian noise variance σ^2 , compare the two posterior probabilities. Their log-likelihood ratio is

$$\log \frac{\mathbb{P}(b = 1 \mid h)}{\mathbb{P}(b = 0 \mid h)} = \log \frac{p}{1-p} + \frac{ah}{\sigma^2} - \frac{a^2}{2\sigma^2}. \quad (4.4)$$

Under equal false-positive and false-negative costs, the optimal threshold is

$$\theta_{\text{Bayes}} = \frac{a}{2} + \frac{\sigma^2}{a} \log \frac{1-p}{p}. \quad (4.5)$$

A rare feature therefore requires stronger evidence before it is declared present. This can reduce total error while increasing the fraction of true rare occurrences that are missed. Report recall as well as average error. Re-optimizing thresholds is an important ablation: the fixed-threshold phase boundary below is exact for its detector family, not a universal Bayes-risk boundary.

4.4 Defining the phase boundary

Let $\mathcal{G}_{\text{mono}}$ and $\mathcal{G}_{\text{super}}$ be two admissible families with the same declared capacity and norm constraints. A population-optimal comparison would use

$$R_a^*(p, I, \sigma, m) = \inf_{G \in \mathcal{G}_a, \theta \in \Theta_a} R(G, \theta; P), \quad a \in \{\text{mono}, \text{super}\}. \quad (4.6)$$

The boundary is $R_{\text{super}}^* = R_{\text{mono}}^*$. This definition is precise, but solving the optimization for all loads and importance schedules is not accomplished merely by writing it. We next solve a fixed, interpretable two-feature family exactly. Learned-model sweeps then test whether the predicted local mechanism appears in trained geometries.

4.5 Distinguishing geometric overlap from statistical interference

The earlier expression $p(n/m) \leq \tau^{-2} - 1$ inserted an average active-feature count into a worst-case alignment formula. An unrestricted attacker can perturb inactive coordinates too; that substitution is not valid. In Equation (4.3), activation probabilities enter the actual distribution of states, so a sparsity-dependent risk has a legitimate derivation.

The project remains a phase-diagram project. Its phase variable is now a defined error comparison rather than an arbitrary cutoff on a scale-dependent proxy.

Status of this result. The finite-state expectation is exact under the detector model. It is a starting formula, not yet a solved global optimization over all learned dictionaries.

Project 1: a closed tradeoff for a superposed pair

Learning objective. Derive a genuine boundary, including rare concepts, importance weights and matched resource controls.

5.1 Two features, one dimension

Take independent binary features with probabilities p_1, p_2 . A superposed code is

$$h = a(b_1 - b_2) + \xi, \quad \xi \sim \mathcal{N}(0, \sigma^2), \quad a > 0. \quad (5.1)$$

Decode with $\hat{b}_1 = \mathbf{1}\{h > a/2\}$ and $\hat{b}_2 = \mathbf{1}\{h < -a/2\}$. This corresponds to a tied matched score with an appropriate threshold. A monosemantic comparator stores feature 1 as $h_m = cb_1 + \xi$, predicts it at threshold $c/2$ and outputs zero for feature 2.

The two encoders have the same latent dimension, but equal $a = c = 1$ gives different squared Frobenius norms: 2 versus 1. Therefore report two controls: equal stored-feature amplitude ($a = c = 1$), and equal total encoder energy ($a = 1/\sqrt{2}, c = 1$). This prevents an apparent robustness effect from being explained entirely by scale.

5.2 Enumerate the four states

Write

$$q = \Phi\left(-\frac{a}{2\sigma}\right), \quad q_3 = \Phi\left(-\frac{3a}{2\sigma}\right), \quad q_m = \Phi\left(-\frac{c}{2\sigma}\right). \quad (5.2)$$

Let $P_{00} = (1 - p_1)(1 - p_2)$, $P_{10} = p_1(1 - p_2)$, $P_{01} = (1 - p_1)p_2$ and $P_{11} = p_1p_2$. Conditional weighted errors are

State	Clean code	Expected weighted detection error
00	0	$(I_1 + I_2)q$
10	a	$I_1q + I_2q_3$
01	$-a$	$I_1q_3 + I_2q$
11	0	$(I_1 + I_2)(1 - q)$

For state 10, missing feature 1 needs $\xi \leq -a/2$, while falsely predicting feature 2 needs $\xi < -3a/2$. For state 11, both features should be present but the decoder can output at most one; each individual feature is missed with probability $1 - q$. This is the nonlinear interference cost.

Theorem 5.1 (Exact pair risk and boundary). *The superposed pair has risk*

$$R_p = (I_1 + I_2)P_{00}q + P_{10}(I_1q + I_2q_3) + P_{01}(I_1q_3 + I_2q) + P_{11}(I_1 + I_2)(1 - q). \quad (5.3)$$

The two zero-output monosemantic comparators have risks

$$R_{m1} = I_1 q_m + I_2 p_2, \quad R_{m2} = I_2 q_m + I_1 p_1. \quad (5.4)$$

For these specified codes and decoders, superposition is preferable exactly when $R_p < \min(R_{m1}, R_{m2})$. Equality defines the boundary.

Proof. Average the four conditional risks using their probabilities. For a stored mono feature, false-positive and false-negative probabilities both equal q_m . The omitted feature is wrong whenever it is active, giving the additional $I_j p_j$ term. $\square$

5.3 The clean boundary in closed form

As $\sigma \rightarrow 0$, $q, q_3, q_m \rightarrow 0$. Thus

$$R_p = (I_1 + I_2) p_1 p_2, \quad R_{m1} = I_2 p_2. \quad (5.5)$$

For $p_2 > 0$, the superposed pair beats the comparator retaining feature 1 precisely when

$$p_1 < \frac{I_2}{I_1 + I_2}. \quad (5.6)$$

For equal frequencies and importances, this is $p < 1/2$, or $s > 1/2$. The mathematical mechanism is explicit: the cost of losing an omitted feature scales with one activation probability, while collision costs scale with a joint activation probability.

If feature 1 is more important, increasing I_1/I_2 makes its protection more valuable and shrinks the region where sharing its direction is worthwhile. For an importance schedule $I_j = j^{-\gamma}$, pairing features i and j gives the corresponding threshold $p_i < j^{-\gamma}/(i^{-\gamma} + j^{-\gamma})$ against retaining i . This is a derived importance dependence, not an inferred slogan that rare concepts are always safe.

5.4 Noise moves the boundary

Equation (5.3) gives an exact boundary in terms of normal CDFs. It need not reduce to a short elementary polynomial to be analytic. Numerical root finding on a proved expression is different from fitting an unknown law to data.

At $p_1 = p_2 = 0.2$, equal importances and equal stored-feature amplitude, the risk crossing is $\sigma \simeq 0.658350$. Under equal total encoder energy, it occurs at $\sigma \simeq 0.279939$. Both comparisons have a tradeoff; its position depends strongly on the resource convention. This is exactly why that convention belongs in the theorem.

5.5 A reconstruction version, to show task dependence

Now let $x_i = b_i U_i$ with uniform amplitudes and noiseless $h = x_1 - x_2$. Decode with ReLUs. When only one feature is active, reconstruction is exact. When both are active, each coordinate's error magnitude is $\min(U_1, U_2)$, and

$$\mathbb{E}[\min(U_1, U_2)^2] = \int_0^1 2t \mathbb{P}(\min(U_1, U_2) > t) dt = \frac{1}{6}. \quad (5.7)$$

Therefore $L_p = (I_1 + I_2) p_1 p_2 / 6$, while dropping feature 2 costs $L_{m1} = I_2 p_2 / 3$. The clean threshold becomes $p_1 < 2I_2 / (I_1 + I_2)$, different from the binary detection threshold. A universal boundary independent of task would miss this difference.

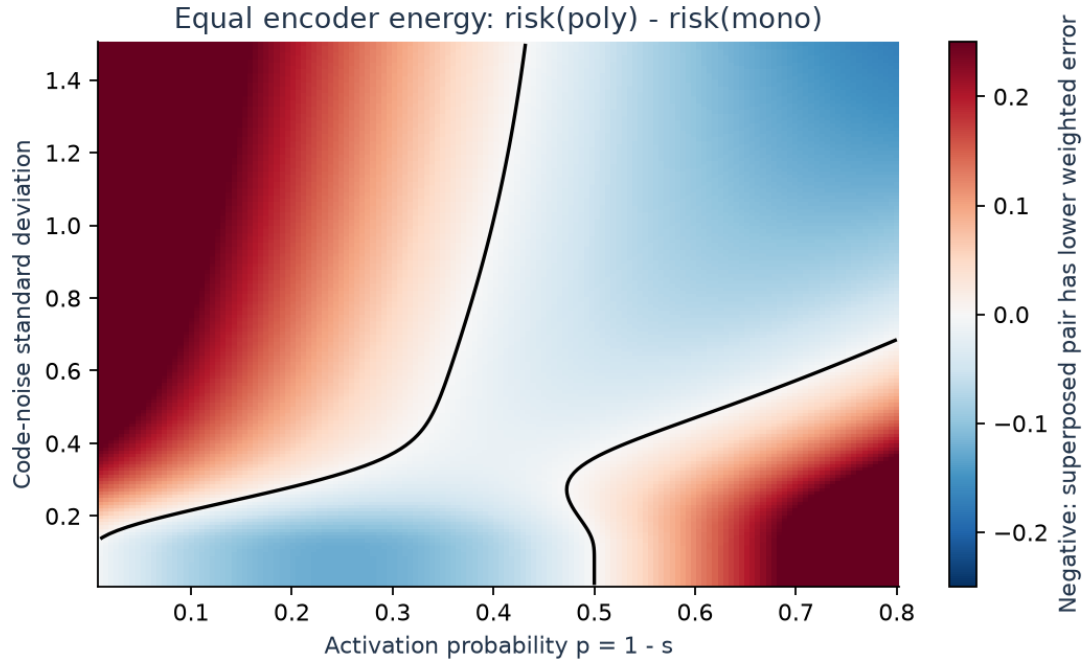


Figure 5.1: Exact equal-energy pair phase diagram. Blue indicates lower weighted error for the superposed pair; red indicates lower error for the specified monosemantic comparator. The black curve is the equality of the two analytic risks. This is not a fitted phase boundary or a result about an arbitrary trained diffusion model.

5.6 What is solved and what is still research

This chapter solves a small, explicit family of codes and decoders. It does not prove that those codes globally minimize the learned autoencoder objective at every parameter setting. The next experiments should compare analytic predictions with measured trained geometries, and test whether optimization changes the boundary. The solved pair is an interpretable unit from which larger constructions and hypotheses can be built.

Status of this result. The risk formulas and pair boundaries are exact under their stated model. Their status as a novel contribution, and their transfer to learned networks, are not established by this derivation alone.

Chapter 6

Project 1: load, interference and a general bound

Learning objective. Connect arbitrary feature load to an error bound without replacing a worst-case norm by an average active count.

6.1 Centering interference

For independent input features, choose

$$\theta_i = \sum_{j \neq i} G_{ij} p_j + \frac{G_{ii}}{2}. \quad (6.1)$$

Then the distance from the conditional mean to threshold is $G_{ii}/2$ for either value of b_i . Define

$$S_i = \sum_{j \neq i} G_{ij} (b_j - p_j), \quad V_i = \sum_{j \neq i} G_{ij}^2 p_j (1 - p_j), \quad B_i = \max_{j \neq i} |G_{ij}|. \quad (6.2)$$

The nuisance fluctuation is $S_i + \zeta_i$, whose variance proxy is $V_i + \sigma^2 G_{ii}$. Unlike M_i , this expression has activation probabilities in a legitimate statistical role.

Theorem 6.1 (A sufficient feature-error bound). *For independent Bernoulli features and independent Gaussian code noise, the detector above satisfies*

$$e_i \leq \exp \left[- \frac{G_{ii}^2}{8 \{ V_i + \sigma^2 G_{ii} + B_i G_{ii} / 6 \}} \right]. \quad (6.3)$$

The bound applies separately to each class-conditional error and hence to their mixture. Degenerate zero-variance cases are interpreted by their deterministic limit.

Proof. Each centered summand $G_{ij}(b_j - p_j)$ has absolute magnitude at most B_i and total variance V_i . Its independent sum obeys the Bernstein exponential-moment bound with variance parameter V_i and scale $B_i/3$. Adding independent Gaussian ζ_i adds $\sigma^2 G_{ii}$ to the variance parameter. Chernoff optimization gives either one-sided tail bound

$$\mathbb{P}(\pm(S_i + \zeta_i) \geq t) \leq \exp \left[- \frac{t^2}{2(V_i + \sigma^2 G_{ii} + B_i t / 3)} \right]. \quad (6.4)$$

Set $t = G_{ii}/2$. The class-0 error uses the upper tail and the class-1 error the lower tail. Each is bounded by the same expression, so averaging them adds no factor two. $\square$

6.2 A robust region with an operational threshold

For a target error $e_i \leq \delta$, a sufficient condition is

$$V_i + \sigma^2 G_{ii} + \frac{B_i G_{ii}}{6} \leq \frac{G_{ii}^2}{8 \log(1/\delta)}. \quad (6.5)$$

This defines a guaranteed region for the specified detector. It is a sufficient bound, not an “if and only if” boundary: failure of the inequality does not imply failure of the detector. The exact pair calculation is sharper in its small setting; the bound covers many features.

6.3 How load enters

For unit columns, define mutual coherence $\mu = \max_{i \neq j} |G_{ij}|$. Then $G_{ii} = 1$, $B_i \leq \mu$ and

$$V_i \leq \mu^2 \sum_{j \neq i} p_j (1 - p_j). \quad (6.6)$$

For independent random unit directions in m dimensions, $\mathbb{E} G_{ij}^2 = 1/m$, so

$$\mathbb{E} V_i = \frac{1}{m} \sum_{j \neq i} p_j (1 - p_j). \quad (6.7)$$

For equal activation probability this is $(n-1)p(1-p)/m$. This is an expectation over random dictionaries, not a universal identity for a trained dictionary or a high-probability guarantee. A concentration argument or direct measured geometry is needed before using it in a theorem about a particular realization.

6.4 A capacity lower bound

For any unit-column dictionary, $\text{tr } G = n$ and $\text{rank } G \leq m$. Cauchy–Schwarz applied to its nonzero eigenvalues gives

$$\text{tr}(G^2) \geq \frac{(\text{tr } G)^2}{m} = \frac{n^2}{m}. \quad (6.8)$$

Subtracting the n diagonal squares and dividing by n yields

$$\frac{1}{n} \sum_i \sum_{j \neq i} G_{ij}^2 \geq \frac{n}{m} - 1. \quad (6.9)$$

This is the familiar frame-potential/Welch-type constraint. It proves unavoidable average overlap when all features have unit norm and $n > m$. It does not upper-bound the worst row, force every pair to overlap, or determine where the optimizer places interference.

6.5 Importance, correlations and learned geometry

Importance enters total risk as $\sum_i I_i e_i$, and also influences W when the training objective is importance-weighted. To predict a boundary purely from $p, n/m, \gamma$, one must characterize the learned geometry as a function of these controls. This remains a substantive theorem target; measuring G and predicting risk conditional on it is a valuable intermediate test, not the same final claim.

If features are correlated, use conditional distributions given b_i and the nuisance covariance. In general, a single centered threshold no longer gives symmetric margins. Under recursive training, initially independent features can become dependent after one generation. Do not keep applying the independent-Bernoulli bound without checking its conditions.

Block-diagonal dictionaries containing isolated columns and antipodal pairs provide an analytically tractable family for loads from one toward two, with risk summed across blocks. They do not cover arbitrary load above two. General state enumeration and the interference bound are the tools for that extension.

Before moving on. Explain the difference between an exact boundary, a sufficient region, an average random-dictionary prediction and a theorem about learned geometry.

Project 1: clean training selects the feature geometry

7.1 The training-to-geometry question

The mathematical task in this chapter is to connect the training objective to the representation that it selects. Once that representation and its decoder are specified, Gaussian feature-detection risk can be evaluated by conditioning on the input states. These are two different steps: predicting error for a supplied code does not establish which code training chooses.

We study the smallest nonlinear setting in which sharing and feature retention can compete. Let b_1, b_2 be independent Bernoulli variables with common activation probability p , and write $q = 1 - p$. There is one latent coordinate. The tied encoder and decoder are

$$\begin{aligned} W &= (u, v), \quad u^2 + v^2 = 1, \quad h = Wb, \\ G &= W^\top W = \begin{pmatrix} u^2 & uv \\ uv & v^2 \end{pmatrix}, \quad \hat{b}_i = \max\{(Gb)_i + \beta_i, 0\}. \end{aligned} \quad (7.1)$$

The total encoder energy is one for every admissible code. Decoder biases β_1, β_2 are free parameters. For positive importance weights, clean population training minimizes

$$L_p(W, \beta) = \sum_{i=1}^2 I_i \sum_{b \in \{0,1\}^2} P_p(b) (\max\{(Gb)_i + \beta_i, 0\} - b_i)^2, \quad (7.2)$$

where the probabilities in state order 00, 01, 10, 11 are q^2, pq, pq, p^2 . This is an exact four-state population objective; no sample approximation or fitted expression enters its definition.

Unless a result explicitly states otherwise, $0 < p \leq 1/2$. General importance weights are retained for the bias and local-allocation calculations. The frequency transition later specializes to $(I_1, I_2) = (1, 1/2)$. This chapter fixes the feature-to-dimension load at two; it does not prove a theorem for arbitrary loads.

7.2 Global decoder-bias minimization at fixed geometry

For a supplied W , the two bias minimizations separate. A positive importance weight multiplies an individual bias objective and therefore does not change its minimizer. It can change the geometry selected when W is optimized jointly.

Fix a feature and abbreviate its four scores by $z_b = (Gb)_i$, its target by $y_b = b_i$, and the state probability by P_b . Its objective is

$$\ell_i(\beta) = \sum_b P_b (\max\{z_b + \beta, 0\} - y_b)^2. \quad (7.3)$$

Its ReLU breakpoints are the distinct numbers $-z_b$. Between consecutive breakpoints the active set $\mathcal{A} = \{b : z_b + \beta > 0\}$ is fixed. Expanding the squared errors gives

$$\begin{aligned}\ell_i(\beta) &= A_0\beta^2 - 2B_0\beta + C_0, \\ A_0 &= \sum_{b \in \mathcal{A}} P_b, \quad B_0 = \sum_{b \in \mathcal{A}} P_b(y_b - z_b), \\ C_0 &= \sum_{b \in \mathcal{A}} P_b(y_b - z_b)^2 + \sum_{b \notin \mathcal{A}} P_b y_b^2.\end{aligned}\tag{7.4}$$

Proposition 7.1 (Finite global bias calculation). *Evaluate every breakpoint and, on every active interval with $A_0 > 0$, the vertex B_0/A_0 clipped to that interval's closure. The least objective value among these candidates is the global fixed-geometry minimum.*

Proof. Within each interval, Equation (7.4) is a quadratic with positive leading coefficient whenever at least one state is active. Its constrained minimum is its vertex if feasible and otherwise an endpoint. An interval with $A_0 = 0$ has constant loss. The far-left inactive interval is represented by its finite boundary; the far-right quadratic is coercive. The original loss is continuous at every breakpoint because a newly active prediction is zero there. Thus the candidates include a minimum on every interval closure and omit no global minimum. $\square$

The complete scalar objective need not be convex. At activation of a state whose target is one, the derivative jumps downward by $2P_b$. This is why identifying one feasible stationary bias is insufficient without comparing the other intervals.

Global sign reversal of W leaves G unchanged. Every geometry can therefore be parametrized, modulo that sign, by

$$W(\theta) = (\cos \theta, \sin \theta), \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad F_{p,I}(\theta) = \sum_i I_i \min_{\beta_i} \ell_i(\beta_i; W(\theta)).\tag{7.5}$$

The clean geometry problem is exactly $\min_{\theta} F_{p,I}(\theta)$. Its reduction to one dimension is exact; a finite angular grid still does not establish its global minimum.

7.3 Monosemantic retention under the same budget

For $W = (1, 0)$, the optimal retained-feature bias is zero and its reconstruction is perfect. The second decoder is a constant. Its optimum is $\beta_2 = p$, since

$$\mathbb{E}[(p - b_2)^2] = p(1 - p) = pq.\tag{7.6}$$

Consequently

$$L_{\text{mono},1} = I_2 pq, \quad L_{\text{mono},2} = I_1 pq.\tag{7.7}$$

The clean-selected comparator retains the more important feature. This choice is made by the training objective, rather than selected afresh at each noise level to obtain a favorable comparison.

Binary detection is a different outcome. With the strict decision $\hat{b}_i > 1/2$, an omitted concept has error p for the declared $p \leq 1/2$, including the tie at $p = 1/2$. A small frequency can therefore make an always-absent detector appear accurate while its recall is zero.

7.4 The equal antipodal pair with trained biases

Consider the supplied equal-energy pair

$$W = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right). \quad (7.8)$$

For either feature, the scores, targets and probabilities can be ordered as

score	0	1/2	-1/2	0
target	0	1	0	1
probability	q^2	pq	pq	p^2

On $0 \leq \beta \leq 1/2$, the negative-score state is inactive and

$$\ell(\beta) = D\beta^2 - 2N\beta + \frac{pq}{4} + p^2, \quad D = 1 - p + p^2, \quad N = \frac{p(1+p)}{2}. \quad (7.9)$$

Proposition 7.2 (Clean-optimal equal-pair bias). *For $0 < p \leq 1/2$, the globally optimal bias and per-feature loss are*

$$\beta_{\text{pair}}^* = \frac{p(1+p)}{2(1-p+p^2)}, \quad \ell_{\text{pair}}^* = \frac{pq}{4} + p^2 - \frac{p^2(1+p)^2}{4D}. \quad (7.10)$$

The weighted loss is $(I_1 + I_2)\ell_{\text{pair}}^*$.

Proof. The nonconstant bias intervals are $(-1/2, 0)$, $(0, 1/2)$ and $(1/2, \infty)$. Their stationary points are respectively $1/2$, N/D and p . For $p \leq 1/2$, $N/D \leq 1/2$, the first point is to the right of its interval, and the last is at or left of its interval. The objective decreases to N/D and then increases. The inactive interval has loss p and cannot improve the displayed solution. Substitution in Equation (7.9) gives the loss. $\square$

For completeness, when $p \geq 1/2$ the same interval comparison selects the all-active bias $\beta = p$ with loss $pq/2$ per feature. The two expressions agree at $p = 1/2$.

The actual trained decoder uses threshold

$$\vartheta = \frac{1}{2} - \beta_{\text{pair}}^* \quad (7.11)$$

on the score before adding the bias. It is not a freely supplied midpoint threshold. For $p = 1/20$, it is approximately 0.4724409449; for $p = 1/5$, it is $5/14$. In particular, rare concepts can have a small positive detection margin even when reconstruction benefits from the pair.

To state the corruption calculation precisely, add Gaussian noise $\sigma\epsilon$ to the latent coordinate, with $\epsilon \sim \mathcal{N}(0, 1)$. The marginal score-noise standard deviation is $\nu = \sigma/\sqrt{2}$. Let Φ denote the standard normal cumulative distribution function. For $\sigma > 0$, the pair's per-feature detection error is

$$\begin{aligned} e_{\text{pair}}(\sigma) &= q^2\Phi(-\vartheta/\nu) + pq\Phi((\vartheta - 1/2)/\nu) \\ &\quad + pq\Phi((-1/2 - \vartheta)/\nu) + p^2\Phi(\vartheta/\nu). \end{aligned} \quad (7.12)$$

Each term is a conditional error for one row of the four-state table. No independence between the two decoder outputs is required to add their weighted marginal risks. At zero noise, evaluate the strict threshold directly rather than divide by $\nu = 0$; the per-feature error is p^2 . For a retained mono concept the error at positive noise is $\Phi(-1/(2\sigma))$, while the omitted concept's error stays p .

7.5 Importance changes the selected amplitudes

For an opposite-sign code, let

$$u = \sqrt{a}, \quad v = -\sqrt{1-a}, \quad r = \sqrt{a(1-a)}. \quad (7.13)$$

Near $a = 1/2$, for $0 < p < 1/2$, both feature decoders have three active states. For a feature with squared column norm d , that branch has

$$\begin{aligned} \beta_3(d) &= \frac{p(1-d+pr)}{D}, \\ \ell_3(d) &= pq(1-d)^2 + p^2(1-d+r)^2 - \frac{p^2(1-d+pr)^2}{D}. \end{aligned} \quad (7.14)$$

The bias lies strictly between its relevant breakpoints at the equal pair when $0 < p < 1/2$. Thus its local differentiation is legitimate. Because $r'(1/2) = 0$, differentiation of $I_1\ell_3(a) + I_2\ell_3(1-a)$ yields

$$\left. \frac{dF}{da} \right|_{a=1/2} = -(I_1 - I_2) \frac{p(1-p)^2(1+p)}{D}. \quad (7.15)$$

If $I_1 > I_2$, moving encoder energy toward concept 1 decreases clean loss. The equal-amplitude pair is therefore not even stationary under unequal importance in this range. This is a statement about training-selected geometry, rather than merely reweighting errors for a fixed code. It does not identify the global optimum at every probability.

7.6 Exact global optima at activation probability one half

Now set $p = 1/2$. Let $a \geq 1/2$ denote the larger squared column norm, $c = 1 - a$ the smaller, and $r = \sqrt{ac}$. The words stronger and weaker in this section refer to column norms, not importance weights.

7.6.1 Bias profiles in both sign sectors

For opposite signs, the strong feature has breakpoints $[-a, r - a, 0, r]$. The stationary points on successive nonempty intervals are

$$c, \quad c + r/2, \quad (2c + r)/3, \quad (c + r)/2. \quad (7.16)$$

The first two lie beyond their interval's right boundary; the third lies in $[0, r]$ because $c \leq r$; the last lies at or left of r . Therefore the third is the global minimum. The weak opposite-sign breakpoints are $[-c, 0, r - c, r]$, with successive roots

$$a, \quad a/2, \quad (2a + r)/3, \quad (a + r)/2. \quad (7.17)$$

The first three lie to the right of their intervals: use $a/2 \geq r - c$, equivalent to $1 + c \geq 2r$, and $a \geq r$. The last is feasible. Hence

$$\beta_s^- = (2c + r)/3, \quad \beta_w^- = (a + r)/2. \quad (7.18)$$

For same signs, the strong breakpoints are $[-a - r, -a, -r, 0]$ and roots

$$c - r, \quad c - r/2, \quad 2(c - r)/3, \quad (c - r)/2. \quad (7.19)$$

The first two are beyond their right boundaries; the third lies in $[-r, 0]$ and the last is to its left. The weak breakpoints are $[-c - r, -r, -c, 0]$ and roots

$$a - r, \quad (a - 2r)/2, \quad 2(a - r)/3, \quad (a - r)/2. \quad (7.20)$$

Again the first three are beyond their right boundaries, using $1 + c \geq 2r$, and the last is feasible. Consequently

$$\beta_s^+ = 2(c - r)/3, \quad \beta_w^+ = (a - r)/2. \quad (7.21)$$

In both sign sectors the globally profiled feature losses are identical:

$$\ell_s = \frac{c^2 + cr + r^2}{6} = \frac{c(1 + r)}{6}, \quad \ell_w = \frac{a}{4}. \quad (7.22)$$

Substitution of the biases into the population quadratic gives these expressions; the identity $c^2 + r^2 = c$ uses $a + c = 1$. Degenerate endpoints follow by direct evaluation or continuity.

7.6.2 Equal importance: unequal norms are globally selected

Theorem 7.3 (Dense equal-importance geometry). *At $p = 1/2$, $I_1 = I_2 = 1$, the global minimum over all energy-one rank-one codes and free biases is*

$$\begin{aligned} a^* &= \frac{1 + 1/\sqrt{2}}{2}, \quad c^* = \frac{1 - 1/\sqrt{2}}{2}, \\ L^* &= \frac{1}{4} - \frac{3 - 2\sqrt{2}}{48} = 0.246425565098879 \dots \end{aligned} \quad (7.23)$$

Both sign sectors and either assignment of the column norms attain it.

Proof. Equations (7.18)–(7.22) profile every sign sector and norm ordering. The total loss is

$$F(a) = \frac{a}{4} + \frac{c(1 + r)}{6} = \frac{1}{4} + \frac{c(2r - 1)}{12}. \quad (7.24)$$

The endpoints $a = 1/2$ and $a = 1$ have loss $1/4$; every interior point has smaller loss. Maximizing $c(1 - 2r)$ gives the stationary equation $c(4a - 1) = r$. Squaring on $1/2 < a < 1$, where both sides are positive, yields

$$(2a - 1)(8a^2 - 8a + 1) = 0. \quad (7.25)$$

The unique interior solution is the stated a^* . Substitution gives the loss, and the endpoint comparison makes it global. $\square$

Equal importance does not force equal encoder-column norms. The result also shows that antipodality is not uniquely required for this dense reconstruction optimum.

7.6.3 Importance ratio two: mono retention is global

Proposition 7.4. *At $p = 1/2$ and $(I_1, I_2) = (1, 1/2)$, the globally optimal geometry retains concept 1 alone, with clean loss $1/8$.*

Proof. If the important concept has the stronger column, the weighted loss is

$$F = \frac{c(1 + r)}{6} + \frac{a}{8} = \frac{1}{8} + c \left(\frac{1 + r}{6} - \frac{1}{8} \right) \geq \frac{1}{8}, \quad (7.26)$$

with equality only at $c = 0$. If it has the weaker column, the loss is $c(1 + r)/12 + a/4 \geq a/4 \geq 1/8$, with a positive additional term at $a = 1/2$. Both sign sectors are covered by Equation (7.22). $\square$

7.7 A computer-assisted global certificate at probability one fifth

Set $p = 1/5$ and $(I_1, I_2) = (1, 1/2)$. Parametrize all finite geometries by

$$W(t) = \frac{(1, t)}{\sqrt{1+t^2}}, \quad t \in \mathbb{R}. \quad (7.27)$$

The point at infinity is retention of concept 2. In the attaining opposite-sign region $-1 < t < 0$, write

$$a = \frac{1}{1+t^2}, \quad c = \frac{t^2}{1+t^2}, \quad r = \frac{-t}{1+t^2}. \quad (7.28)$$

The important feature has three active states; the weaker feature has only states 00, 01 active. Minimizing their population quadratics gives

$$\begin{aligned} \beta_1 &= \frac{p(c+pr)}{D}, & L_1 &= pqc^2 + p^2(c+r)^2 - \frac{p^2(c+pr)^2}{D}, \\ \beta_2 &= pa, & L_2 &= pq^2a^2 + p^2. \end{aligned} \quad (7.29)$$

The weak active set is feasible when $pa \leq r - c$, or

$$t^2 + t + \frac{1}{5} \leq 0, \quad \frac{-1 - 1/\sqrt{5}}{2} \leq t \leq \frac{-1 + 1/\sqrt{5}}{2}. \quad (7.30)$$

The strong bias is feasible on this interval. Substitution into $L_1 + L_2/2$ gives

$$F(t) = \frac{905t^4 - 320t^3 + 410t^2 + 441}{5250(1+t^2)^2}, \quad (7.31)$$

$$F'(t) = \frac{8t(20t^3 + 175t^2 - 60t - 59)}{2625(1+t^2)^3}. \quad (7.32)$$

Theorem 7.5 (Certified low-frequency optimum). *The global clean optimum in this declared case is $W(t^*)$, where*

$$\begin{aligned} 20t^{*3} + 175t^{*2} - 60t^* - 59 &= 0, \\ -0.442090239023400245 < t^* < -0.442090239023400243. \end{aligned} \quad (7.33)$$

Its biases and loss are

$$\begin{aligned} \beta_1^* &= \frac{t^*(5t^* - 1)}{21(1+t^{*2})}, \quad \beta_2^* = \frac{1}{5(1+t^{*2})}, \\ 0.07775208251629429 < L^* < 0.07775208251629430. \end{aligned} \quad (7.34)$$

The global comparison is a finite exact-rational computer-assisted certificate, in addition to the explicit attaining-branch derivation.

Proof. Proposition 7.1 supplies four breakpoints and four feasible active-interval vertices for each feature at a generic geometry. The inactive constant is represented by the first breakpoint. Score order changes only at $t = -1, 0, 1$, so four regions each have $8 \times 8 = 64$ joint bias candidates. Their losses are rational functions with positive denominators proportional to $(1+t^2)^2$; bias feasibility gives polynomial inequalities of degree at most two. Their stationary equations have degree at most four.

On a feasible interval a candidate function attains its finite minimum at a stationary point or a feasibility or region endpoint. Infinite endpoints are covered by the separately solved mono

geometry. Thus isolating these finitely many roots and comparing their objective values covers every geometry and bias minimum; a local angular optimum or a numerical grid is not used as the certificate.

The exact-rational implementation enumerated 256 joint branches and 1,031 raw root candidates. After feasibility checks and duplicate removal, 265 distinct function/root candidates remained. Feasibility signs were proved by rational interval evaluation and polynomial greatest common divisors for exact shared roots. Objective values were enclosed by rational interval arithmetic. The upper enclosure for the stated winner lies below every competing lower enclosure. The smallest such competitor lower bound exceeds 0.07827150837086602; mono retention losses are 0.08 and 0.16. No competitor enclosure overlaps the winner. The displayed root is strictly inside its active-set feasibility interval. This proves the global claim. $\square$

The competing regional minimum values, rounded only for exposition, are 0.0942857143 for $t \leq -1$, 0.0777520825 for $-1 \leq t \leq 0$, 0.08 for $0 \leq t \leq 1$, and 0.1569262176 for $t \geq 1$. The certificate compares rational enclosures rather than these decimals.

The code and full candidate ledger are preserved in `project1_toy/focused_bridge_2026-10-06/math/certify_low_p.py` and `project1_toy/focused_bridge_2026-10-06/math/low_p_certificate_candidates.json`. The summary contains the exact root and objective enclosures. The independent review reproduced the winner using a different active-subset enumeration.

Despite its lower clean squared error, the selected weaker feature never passes the actual decoder's detection threshold:

$$\max_b \hat{b}_2 = c + \beta_2 = \frac{t^{*2} + 1/5}{1 + t^{*2}} < \frac{1}{2}. \quad (7.35)$$

The important feature is detected correctly in all four clean states; the weighted clean detection error is therefore $\frac{1}{2}p = 0.1$, equal to mono's clean detection error. Reconstruction improvement cannot be substituted for feature-detection improvement.

7.8 The global frequency transition at importance ratio two

Fix $(I_1, I_2) = (1, 1/2)$ while allowing $0 < p \leq 1/2$.

Theorem 7.6 (Clean geometry-selection transition). *Let*

$$p_c = \frac{3 - \sqrt{5}}{2} = 0.3819660112501051518\dots \quad (7.36)$$

For $p_c \leq p \leq 1/2$, retention of concept 1 alone is the globally optimal geometry, with biases $(0, p)$ and loss $pq/2$. Every mixed code has strictly greater loss. For $0 < p < p_c$, a mixed opposite-sign code strictly improves on mono, so every global minimizer has both columns nonzero. This theorem does not assert a unique explicit sharing geometry at every $p < p_c$.

The proof requires global comparison, not merely a change of local stability. We give the complete bias profiles and then compare their losses. Let the larger squared norm be $a \geq 1/2$, set $c = 1 - a$, $r = \sqrt{ac}$, and introduce

$$k = \sqrt{c/a} \in [0, 1], \quad a = (1 + k^2)^{-1}, \quad c = k^2(1 + k^2)^{-1}, \quad r = k(1 + k^2)^{-1}. \quad (7.37)$$

7.8.1 Excluding all same-sign sharing

The strong feature has breakpoints $[-a-r, -a, -r, 0]$. Its successive stationary biases at general p are

$$c-r, \quad c-pr, \quad \frac{c-r}{2-p}, \quad p(c-r). \quad (7.38)$$

The first two lie past the right boundaries, the third lies in $[-r, 0]$, and the last is to its left. The global strong-feature solution is

$$\beta_s^+ = \frac{c-r}{2-p}, \quad S_+ = \frac{pq c(1+2qr)}{2-p}. \quad (7.39)$$

For the weak feature, any bias $\beta \leq -c$ misses its single-feature state and has loss at least pq . The all-active candidate $\beta_w^+ = p(a-r) \geq 0$ has loss $W_+ = pqa \leq pq$. For $\beta \geq -c$, the three-active vertex $(a-r)/(2-p)$ lies at or beyond zero, so that interval decreases toward the all-active interval. The global weak solution is therefore

$$\beta_w^+ = p(a-r), \quad W_+ = pqa. \quad (7.40)$$

This comparison covers possible inferior negative-bias local minima rather than discarding them by a convexity assumption.

Since $2r \leq 1$, $S_+ \leq pqc \leq W_+$. Assigning the larger importance to the stronger column is therefore the better orientation. Even it has

$$S_+ + \frac{1}{2}W_+ - \frac{1}{2}pq = \frac{pq c(p/2 + 2qr)}{2-p} > 0 \quad (7.41)$$

for every mixed code. Thus same-sign sharing never improves mono in this importance setting, for the full declared probability range.

7.8.2 Opposite-sign strong-feature profile

The strong feature's breakpoints are $[-a, r-a, 0, r]$. The interval roots are

$$c, \quad c+pr, \quad \frac{p(c+pr)}{D}, \quad p(c+r). \quad (7.42)$$

The first two lie to the right of their intervals. The third lies in $[0, r]$ because $pc \leq qr$ follows from $p \leq q$ and $c \leq r$. The last root lies at or left of r by the same inequality. Hence

$$\begin{aligned} \beta_s^- &= \frac{p(c+pr)}{D}, \\ S_- &= pqc - \frac{pq}{D}(qr-pc)^2 = \frac{pqc}{D} [p + (1-2p)c + 2pqr]. \end{aligned} \quad (7.43)$$

The second equality follows by expanding and using $r^2 = ac$, $a+c=1$.

7.8.3 Every competing weak-feature bias branch

The weak feature's breakpoints are $[-c, 0, r-c, r]$. All possible global minima are among the following feasible candidates:

$$\beta_{w,2} = pa, \quad W_2 = pq^2a^2 + p^2, \quad (7.44)$$

$$\beta_{w,3} = \frac{p(a+pr)}{D}, \quad W_3 = pqa - \frac{pq}{D}(qr-pa)^2, \quad (7.45)$$

$$\beta_{w,4} = p(a+r), \quad W_4 = pqa. \quad (7.46)$$

The two-active branch in Equation (7.44) is feasible when $p \leq k - k^2$; the all-active branch in Equation (7.46) is feasible when $k \leq p/q$. The three-active branch is feasible when $p + Dk^2 - qk \geq 0$ and $k \geq p/q$. It also equals

$$W_3 = \frac{pqa}{D} [p + (1 - 2p)a + 2pqr]. \quad (7.47)$$

Here is why the candidate list is complete. The first interval's single target-one stationary bias lies beyond its right boundary. The two-active quadratic has vertex pa ; the three-active and all-active vertices are as displayed. A target-one activation has a strictly downward derivative jump and cannot be a minimum at its kink. Target-zero activation has a continuous derivative and its possible minimum is covered by adjacent interval vertices on their closures. The far-left inactive loss is p ; for $c > 0$, the feasible bias zero has loss $pqa^2 + p^2 < p$, and for a dropped column the optimal constant has loss $pq < p$. Thus the constant cannot win. At low probabilities two- and three-active minima can coexist; taking only one of them without comparison would be unjustified.

For the upper-phase proof, $p \geq p_c > 1/4$. Equation (7.44) is then impossible, since $k - k^2 \leq 1/4$. For $k \geq p/q$, the remaining three-active lower-feasibility condition follows from

$$pa + Dc - qr = [pa - (r - c)] + p(r - qc) > 0. \quad (7.48)$$

The first bracket is positive because the excluded two-active branch is infeasible; the second is nonnegative. Therefore above p_c the weak profile is precisely W_4 for $k \leq p/q$ and W_3 for $k \geq p/q$.

Both profiles are at least the strong loss. For W_4 , use $S_- \leq pqc \leq pqa$. For W_3 , subtraction gives

$$W_3 - S_- = \frac{pq(a - c)}{D} [q + 2pqr] \geq 0. \quad (7.49)$$

Thus assigning importance one to the stronger column is the better orientation in the upper phase; the reverse assignment is covered without assuming it away.

7.8.4 Global comparison above the threshold

In the all-active weak branch, the mixed-code difference from mono is

$$\Delta_A = S_- + \frac{1}{2}pqa - \frac{1}{2}pq = \frac{pqc}{D} \left[p - \frac{D}{2} + (1 - 2p)c + 2pqr \right]. \quad (7.50)$$

For $p_c \leq p \leq 1/2$, all bracketed terms are nonnegative because $p - D/2 = -(p^2 - 3p + 1)/2 \geq 0$. For every mixed code, the bracket is strictly positive, including at $p = p_c$.

For the three-active weak branch, direct substitution gives

$$\Delta_3 = \frac{pq}{2D(1 + k^2)^2} H_p(k), \quad (7.51)$$

$$H_p(k) = (1 - p - p^2)k^4 + 4pqk^3 + (-2 + 5p - 2p^2)k^2 + 2pqk - p^2. \quad (7.52)$$

Set $A = 1 - p - p^2$ and $B = -2 + 5p - 2p^2$. On $[p_c, 1/2]$, $A \geq 1/4$ and $B + p = -2(p^2 - 3p + 1) \geq 0$. For $k \geq p/q$,

$$H'_p(k) = 4Ak^3 + 2pk(6qk - 1) + 2pq + 2(B + p)k > 0, \quad (7.53)$$

because $qk \geq p \geq p_c > 1/6$. At $k = p/q$, the weak branches agree and H_p is positive by Equation (7.50). It remains positive throughout the three-active branch. This covers every mixed opposite-sign code above the threshold. Same-sign codes and importance swaps have already been covered, while retention of the unimportant concept has loss $pq > pq/2$. Mono is therefore globally optimal, not merely locally stable.

7.8.5 Strict sharing improvement below the threshold

For a sufficiently small opposite-sign perturbation of important-concept mono, $k > 0$ lies in the weak all-active branch. Since $c = k^2 + O(k^4)$ and $r = k + O(k^3)$, Equation (7.50) gives

$$\Delta_A = pq \left(\frac{p}{D} - \frac{1}{2} \right) k^2 + O(k^3). \quad (7.54)$$

For $p < p_c$ the leading coefficient is strictly negative, so a sufficiently small mixed code has lower clean loss than mono. A global optimum exists: the energy-one code circle is compact and score magnitudes are at most $\sqrt{2}$; a minimizing bias can be restricted to $[-3, 3]$ without losing an optimum. Loss is continuous on this compact parameter set. Therefore the global optimum cannot be mono below p_c , and cannot be same-sign by Equation (7.41). This completes the proof of Theorem 7.6.

7.9 What is established and what remains a research question

The clean allocation threshold is global: a competing off-axis optimum does not preempt the local stability threshold. In the sparsity convention $s = 1 - p$, sharing is selected for $s > (\sqrt{5} - 1)/2$, while important-concept retention is selected at and below that sparsity value.

This is a clean-training phase, not automatically a robustness phase. Corruption risk must be calculated using the selected W and its actual trained biases. The exact fixed-state Gaussian expression

$$e_i(\sigma) = \sum_b P_p(b) \Phi \left[-\frac{(2b_i - 1)((Gb)_i + \beta_i - 1/2)}{\sigma \sqrt{G_{ii}}} \right], \quad \sigma > 0, \quad (7.55)$$

applies to stored features; a dropped feature is evaluated as a constant decision. The following corruption analysis uses this distinction to connect selected geometry to operational errors rather than infer robustness from a clean reconstruction label.

The finite-bias and dense-case formulas were independently reviewed. The low-frequency certificate used exact root isolation and received a separate independent active-subset check. The frequency proof's identities are recorded in `project1_toy/frequency_boundary_2026-10-06/math/verification_results.json`; an independent review checked the population quadratics and branch coverage. These checks substantiate the declared toy mathematics. They do not by themselves certify research novelty, prove a variable-load law, demonstrate real-model transfer, or guarantee conference acceptance.

Project 1: robustness of the clean-selected representation

Learning objective. Evaluate the decoder selected by clean reconstruction, and prove exactly what its noise comparison establishes.

8.1 A fixed evaluation rule after clean training

Let the independent bits b_1, b_2 have common activation probability p , put $q = 1 - p$, and retain the encoder-energy constraint $\|W\|_F^2 = 1$. The clean objective is weighted squared reconstruction with weights $(1, 1/2)$. Once W and β have been selected by that objective, they remain fixed during noise evaluation. For scalar code noise $\xi \sim \mathcal{N}(0, \sigma^2)$, independent of the bits,

$$h = Wb + \xi, \quad \hat{b}_i^{\text{amp}} = \text{ReLU}((Gb)_i + \beta_i + w_i \xi), \quad G = W^\top W. \quad (8.1)$$

The evaluated binary decision is the actual trained decoder's strict rule

$$\hat{b}_i = \mathbf{1}\{\hat{b}_i^{\text{amp}} > 1/2\} = \mathbf{1}\{(Gb)_i + \beta_i + w_i \xi > 1/2\}. \quad (8.2)$$

The equality follows because the threshold is positive: ReLU exceeds $1/2$ exactly when its preactivation does. The weighted detection error is

$$R(W, \beta; p, \sigma) = r_1 + \frac{1}{2}r_2, \quad r_i = \mathbb{P}\{\hat{b}_i \neq b_i\}. \quad (8.3)$$

Neither the detector nor the mono comparator is reselected to improve noisy performance. This is average Gaussian code-noise robustness of a specified decision rule; it is not a worst-case input-attack theorem, a Bayes-optimal detector comparison, or an AI-safety evaluation.

8.2 Deriving the exact four-state Gaussian risk

Define the clean threshold gap $m_i(b) = (Gb)_i + \beta_i - 1/2$. Conditional on b , $w_i \xi$ has zero mean and variance $\sigma^2 w_i^2 = \sigma^2 G_{ii}$. If $w_i \neq 0$ and $\sigma > 0$, the two conditional errors are

$$\mathbb{P}(\hat{b}_i = 1 \mid b_i = 0, b_{-i}) = \Phi\left(\frac{m_i(b)}{\sigma|w_i|}\right), \quad (8.4)$$

$$\mathbb{P}(\hat{b}_i = 0 \mid b_i = 1, b_{-i}) = \Phi\left(\frac{-m_i(b)}{\sigma|w_i|}\right). \quad (8.5)$$

For the first identity, use $\mathbb{P}(\xi' > -m) = \Phi(m/s)$ for $\xi' \sim \mathcal{N}(0, s^2)$. For the second, use $\mathbb{P}(\xi' \leq -m) = \Phi(-m/s)$. Both are captured by

$$r_i(p, \sigma) = \sum_{b \in \{0,1\}^2} P_p(b) \Phi\left(\frac{(1 - 2b_i)m_i(b)}{\sigma|w_i|}\right). \quad (8.6)$$

The two output errors can be correlated through their shared scalar noise. Independence of those errors is unnecessary: linearity of expectation gives $R = r_1 + r_2/2$ regardless. For $w_i = 0$ or $\sigma = 0$, evaluate the deterministic decision directly instead of dividing by zero.

For an opposite-sign encoder $W = (1, t)/\sqrt{1 + t^2}$ with $t < 0$, write $a = 1/(1 + t^2)$, $c = t^2/(1 + t^2)$, and $d = t/(1 + t^2) < 0$. The complete state enumeration is

State b	Probability	$m_1(b)$	$m_2(b)$
00	q^2	$\beta_1 - 1/2$	$\beta_2 - 1/2$
10	pq	$a + \beta_1 - 1/2$	$d + \beta_2 - 1/2$
01	pq	$d + \beta_1 - 1/2$	$c + \beta_2 - 1/2$
11	p^2	$a + d + \beta_1 - 1/2$	$c + d + \beta_2 - 1/2$

This table supplies all eight state/feature CDF terms. It makes explicit how negative overlap can suppress false positives while simultaneously cancelling a coactive concept's positive signal.

8.3 The clean-selected mono control

For $0 < p \leq 1/2$, mono retention of the important concept is $W = (1, 0)$ and $\beta = (0, p)$. The retained concept's threshold gaps are $-1/2$ and $+1/2$, so its error is $\Phi(-1/(2\sigma))$ for either class. The omitted concept has the constant amplitude prediction p and is never declared present under the strict rule, including the tie at $p = 1/2$. Consequently

$$\boxed{R_{\text{mono}}(p, \sigma) = \Phi\left(-\frac{1}{2\sigma}\right) + \frac{p}{2}, \quad \sigma > 0,} \quad R_{\text{mono}}(p, 0) = \frac{p}{2}. \quad (8.7)$$

The retained-feature error's lack of dependence on p is a property of its supplied symmetric threshold gaps. A detector optimized separately for rare-class detection would generally use a different threshold.

8.4 Why reconstruction and detection can disagree

At $p = 1/5$, the globally certified clean optimum has $t_* \approx -0.4420902390$ and weighted reconstruction loss 0.0777520825163, below mono's 0.08. Its weak concept's maximum reconstruction is

$$c + \beta_2 = \frac{t_*^2 + 1/5}{1 + t_*^2} < \frac{1}{2}. \quad (8.8)$$

The inequality is checked at the isolated algebraic root, not inferred from a rounded angle. Thus the weak concept is never detected, even when present alone. The important concept is detected correctly in every clean state, and the total clean detection error equals mono's $p/2 = 0.1$. A nonzero stored column and improved amplitude estimation therefore do not imply improved concept detection. All primary positive-noise settings tested at this frequency yield greater detection error for sharing than for mono.

8.5 The infinite-noise comparison

Let the encoder be genuinely shared, meaning both $w_i \neq 0$, and keep its finite biases fixed. Every argument in Equation (8.6) tends to zero as $\sigma \rightarrow \infty$. Since $\Phi(0) = 1/2$ and the state

probabilities sum to one, $r_i \rightarrow 1/2$. Therefore

$$R_{\text{shared}}(p, \sigma) \rightarrow \frac{3}{4}, \quad (8.9)$$

$$R_{\text{mono}}(p, \sigma) \rightarrow \frac{1}{2} + \frac{p}{2}, \quad (8.10)$$

$$\boxed{\Delta(p, \sigma) := R_{\text{shared}} - R_{\text{mono}} \rightarrow \frac{1}{4} - \frac{p}{2} > 0} \quad (0 < p < 1/2). \quad (8.11)$$

A sharing advantage at a finite noise level cannot persist to arbitrarily large noise under this fixed rule. The omitted mono feature receives no code noise, whereas every nonzero shared column eventually randomizes. This asymptotic conclusion does not assert which finite noise levels favor sharing.

To derive the possible noncommutation of limits explicitly, fix $p < 1/2$ and consider the supplied family

$$W_\varepsilon = (\sqrt{1 - \varepsilon^2}, -\varepsilon), \quad \beta = (0, p), \quad 0 < \varepsilon < 1. \quad (8.12)$$

At any fixed finite $\sigma > 0$, the weak column's score and noise tend to zero as $\varepsilon \downarrow 0$, leaving a constant prediction $p < 1/2$ and weak error p . The strong column converges to mono's Gaussian detector. Hence $R_\varepsilon(p, \sigma) \rightarrow R_{\text{mono}}(p, \sigma)$. For every fixed $\varepsilon > 0$, both columns are nonzero, so $R_\varepsilon(p, \sigma) \rightarrow 3/4$ as $\sigma \rightarrow \infty$. It follows that

$$\lim_{\sigma \rightarrow \infty} \lim_{\varepsilon \downarrow 0} R_\varepsilon = \frac{1}{2} + \frac{p}{2} \neq \frac{3}{4} = \lim_{\varepsilon \downarrow 0} \lim_{\sigma \rightarrow \infty} R_\varepsilon. \quad (8.13)$$

This proves the limit distinction for a supplied family; it does not claim that every member is clean-optimal.

8.6 A rigorous example with at least two crossings

The fixed $p = 1/20$ case has an exactly certified global clean-selected mixed geometry. Its parameter is the unique root in the rational interval

$$\left[-\frac{591234926798208727}{913174160928329335}, -\frac{93541350199299932}{144476484910883523} \right] \quad (8.14)$$

of

$$380t^3 + 14800t^2 - 1140t - 6839 = 0, \quad t_* \approx -0.6474503463798862. \quad (8.15)$$

The attaining branch has biases

$$\beta_1 = \frac{20t_*^2 - t_*}{381(1 + t_*^2)}, \quad \beta_2 = \frac{1}{20(1 + t_*^2)}. \quad (8.16)$$

These follow from the three-active strong-feature and two-active weak-feature population quadratics. Global selection follows from the finite-branch certificate described in the preceding chapter; the cubic alone would establish only a stationary candidate.

To obtain the cubic explicitly, substitute $p = 1/20$, $a = 1/(1 + t^2)$, $c = t^2/(1 + t^2)$, $r = -t/(1 + t^2)$ and $D = 381/400$ into the attaining branch in Equation (7.29). Its weighted clean objective and derivative simplify to

$$F_{1/20}(t) = \frac{296420t^4 - 28880t^3 + 30440t^2 + 145161}{6096000(1 + t^2)^2}, \quad (8.17)$$

$$F'_{1/20}(t) = \frac{19t(380t^3 + 14800t^2 - 1140t - 6839)}{1524000(1 + t^2)^3}. \quad (8.18)$$

The weak two-active branch requires $p \leq -t - t^2$, and the isolated winner is strictly feasible. The same complete region/bias/root comparison used for $p = 1/5$ certifies this winner against every other branch and both mono endpoints. Thus differentiation explains the polynomial, while the separate global comparison explains why its specified root is selected.

Theorem 8.1 (Two-crossing existence for the certified rare-concept code). *For this fixed geometry, the actual-decoder weighted detection-risk difference from mono obeys*

$$\Delta(0) = \frac{1}{400} > 0, \quad \Delta(3/10) < 0, \quad \lim_{\sigma \rightarrow \infty} \Delta(\sigma) = \frac{9}{40} > 0. \quad (8.19)$$

It has at least one zero in $(0, 3/10)$ and a distinct zero in $(3/10, \infty)$.

Proof. The eight clean gaps in the state table are strictly nonzero at the root enclosure. The important concept errs only in state 11, whose probability is $p^2 = 1/400$. The weak concept is never detected, contributing weighted error $p/2 = 1/40$. Hence $R_{\text{shared}}(p, 0) = 11/400$, while $R_{\text{mono}}(p, 0) = 1/40$, and their difference is $1/400$.

At $\sigma = 3/10$, the rational CDF enclosure derived below proves a strictly negative difference. Equation (8.11) gives the limiting difference $1/4 - (1/20)/2 = 9/40$. For positive σ , the error is a finite sum of continuous CDF terms. Strict clean gaps imply that each term tends to its deterministic clean error as $\sigma \downarrow 0$, so the positive clean limit agrees with $\Delta(0)$. There is therefore a positive point below $3/10$ with positive difference and a finite point above $3/10$ with positive difference. Apply the intermediate value theorem on the two disjoint intervals with the negative value at $3/10$ as their shared endpoint. $\square$

The theorem proves existence, not exactly two roots, their locations, or a result for every frequency. In particular, a single permanent “robust phase” after one noise crossing is not an adequate description of this example.

8.7 Deriving the finite-noise rational certificate

8.7.1 From an algebraic root to interval probabilities

Represent the isolated root by a rational interval $T = [\ell, u]$. Evaluate $1 + t^2$, the Gram entries, biases, and nonzero column norms by outward interval arithmetic. For intervals $A = [a, b]$, $B = [c, d]$,

$$A + B = [a + c, b + d], \quad -A = [-b, -a], \quad (8.20)$$

$$AB = [\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)]. \quad (8.21)$$

If $0 \notin B$, reciprocation maps its endpoints to $B^{-1} = [1/d, 1/c]$, and division uses multiplication by this interval. Repeated occurrences of t can widen the result but cannot invalidate containment.

For a nonnegative rational x and $S = 10^{30}$, let

$$n = \left\lfloor \sqrt{xS^2} \right\rfloor. \quad (8.22)$$

Integer square root applied to $\lfloor xS^2 \rfloor$ computes n exactly. Then $n/S \leq \sqrt{x} < (n+1)/S$, with coincident endpoints allowed for an exact grid square. Apply monotonicity at both endpoints of a nonnegative interval to enclose its square root. This gives rational bounds for $|w_i|$ and $1/\sqrt{2\pi}$ without relying on floating-point square roots.

For each state and feature, the resulting interval encloses

$$x_{i,b} = \frac{(1 - 2b_i)m_i(b)}{(3/10)|w_i|}. \quad (8.23)$$

Round its lower endpoint downward and upper endpoint upward to a 10^{-12} grid. Monotonicity of Φ reduces the remaining task to bounding Φ at two rational endpoints.

8.7.2 An exact enclosure of the normalizing constant

For $0 < x < 1$, write

$$A_{30}(x) = \sum_{j=0}^{30} \frac{(-1)^j x^{2j+1}}{2j+1}, \quad A_{31}(x) = A_{30}(x) - \frac{x^{63}}{63}. \quad (8.24)$$

The alternating-series remainder gives $A_{31}(x) \leq \arctan x \leq A_{30}(x)$. Machin's identity is

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239). \quad (8.25)$$

To justify its quadrant as well as its tangent, doubling $\tan \alpha = 1/5$ gives $\tan(2\alpha) = 5/12$ and $\tan(4\alpha) = 120/119$. Hence

$$\tan(4\alpha - \arctan(1/239)) = \frac{120/119 - 1/239}{1 + 120/(119 \cdot 239)} = 1. \quad (8.26)$$

Also $x/(1+x^2) < \arctan x < x$ for $x > 0$, by the integral definition. Thus the angle is positive and below $4/5$. Since $\arctan 1 > 1/2$, $\pi/2 = 2 \arctan 1 > 1 > 4/5$. The angle lies in $(0, \pi/2)$ and must equal $\pi/4$. Combining the rational arctangent bounds encloses π ; monotone square root and reciprocation then enclose $1/\sqrt{2\pi}$.

8.7.3 Rational upper and lower Gaussian CDF bounds

For $x \geq 0$,

$$\Phi(x) = \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_0^x e^{-u^2/2} du. \quad (8.27)$$

Taylor's theorem at zero for e^{-v} , $v \geq 0$, gives

$$e^{-v} = \sum_{j=0}^N \frac{(-v)^j}{j!} + \frac{(-1)^{N+1} e^{-\eta}}{(N+1)!} v^{N+1}, \quad 0 \leq \eta \leq v. \quad (8.28)$$

An even N therefore gives an upper bound and the next odd degree a lower bound, regardless of whether the initial term magnitudes decrease. Substituting $v = u^2/2$ and integrating term by term yields the rational polynomial

$$J_N(x) = \sum_{j=0}^N \frac{(-1)^j x^{2j+1}}{2^j j! (2j+1)}, \quad J_{81}(x) \leq \int_0^x e^{-u^2/2} du \leq J_{80}(x). \quad (8.29)$$

Combine this interval with the normalizing-constant interval. Negative arguments use $\Phi(-x) = 1 - \Phi(x)$; intersecting a valid enclosure with $[0, 1]$ preserves validity. Every operation uses rational arithmetic.

8.7.4 The certified sign and what is independently checked

Multiply each enclosed CDF by the exact state probability and feature weight and sum the eight contributions. Evaluate the mono term at argument $-5/3$ with the same CDF enclosure and add $p/2$. Interval subtraction yields a wholly negative enclosure for $\Delta(3/10)$, with displayed approximate endpoints

$$[-0.014474506712828471, -0.014474506712749220]. \quad (8.30)$$

The decimal endpoints summarize exact rational endpoints saved in the certificate; the sign decision is made using those rational endpoints. An independent 100-digit reference value is approximately $-0.0144745067127706407821$ and lies inside the enclosure. That reference is a numerical crosscheck, not the sign proof.

Independent review checked the root and bias mapping, all eight strict clean gaps, the outward operations, Taylor and arctangent recurrences, and the continuity argument. Fresh reproduction in a separate directory matched the saved certificate except runtime. The full rational per-state records are in `project1_toy/frequency_boundary_2026-10-06/math/noise_sign_certificate_results.json`; the implementation is `certify_noise_sign.py` in that directory.

8.8 The fixed numerical checks and their interpretation

Five frequencies were declared before the verification: .05, .20, .35, .375, .40. The first four use global population certificates; the last uses the analytic mono theorem. Selected clean losses and weak-column energies are

p	Weak-column energy	Selected MSE	Mono MSE
.05	.295373681287	.017742880541	.023750000000
.20	.163490565429	.077752082516	.080000000000
.35	.002595127216	.113740582191	.113750000000
.375	.000121567823	.117187402718	.117187500000
.40	0	.120000000000	.120000000000

These are discrete certified selections, not a sampled proof of the continuous transition. Near-transition MSE gains are small: approximately 9.42×10^{-6} at .35 and 9.73×10^{-8} at .375.

The original primary noise settings remain 0, .05, .15, .30, .60. The actual-decoder selected-minus-mono risk differences are

p	0	.05	.15	.30	.60
.05	.002500	.002500	.003472	-.014475	-.003661
.20	0	.012228	.017397	.011652	.007769
.35	0	≈ 0	.000053	-.000144	-.000863
.375	0	≈ 0	-.000001	-.000135	-.000223
.40	0	0	0	0	0

Negative values favor sharing. Approximate zeros can reflect tiny Gaussian tails or finite-precision CDF saturation, so they do not certify exact equality. At .40 the selected geometry is exactly mono, so equality holds independently of floating-point evaluation.

A prespecified 240-point logarithmic display grid from noise .001 to 4 and Brent refinement of reliable adjacent sign changes gave numerical roots

p	Lower crossing	Upper crossing
.05	.2079551941	.6252753374
.35	.2698862926	1.0932765901
.375	.1277396994	3.3971473354

No crossing was found for .20 in the display interval. This is not a proof of no crossing at all noise strengths, nor a complete root count for the other cases. The separate rational theorem proves existence of at least two roots only at .05.

8.9 Verification failures and stopping scope

The preceding projected-Adam campaign preserved all 36 final iterates. Twenty-seven passed its loss-stability criterion and nine failed; passing loss stability did not ensure stationarity, and several iterates retained substantial tangent gradients. The separate two-case angular solver reached gradient tolerance at the already specified .05 and .20 unequal-importance cases. Later global certificates establish population optima, not a guarantee that every solver or initialization learns them. The frozen training files were not replaced by the certificate outputs.

The independent numerical review checked 48 raw hashes, 150 primary risk rows, 1,200 display values, and six numerical root brackets. It recovered the four sharing optima using a different 225-active-subset construction, while explicitly sharing the reviewed rational-root backend. Maximum detection and display discrepancies were below 1.4×10^{-16} .

The initial independent MSE checker failed because unbounded quadrature missed central Gaussian mass when a weak-feature ReLU cutoff lay far into a tail. Its source and discrepancies remain archived. The repaired checker integrates over $[-12, 12]$, splitting at zero and any relevant ReLU cutoff, and bounds the omitted tails. If a conditional preactivation is $\mu + sZ$ and the target bit is b , then

$$|\text{ReLU}(\mu + sZ) - b|^2 \leq 2(|\mu| + 1)^2 + 2s^2Z^2. \quad (8.31)$$

For $A > 0$, Mills' bound and integration by parts give

$$Q(A) \leq \frac{\phi(A)}{A}, \quad \mathbb{E}[Z^2 \mathbf{1}\{|Z| > A\}] = 2\{A\phi(A) + Q(A)\}. \quad (8.32)$$

Therefore the omitted two-sided contribution is bounded by

$$4(|\mu| + 1)^2 Q(A) + 4s^2\{A\phi(A) + Q(A)\}. \quad (8.33)$$

At $A = 12$, the largest bound over the reviewed cases was below 3.90×10^{-31} ; corrected quadrature agreed with saved MSE values to 1.39×10^{-16} . Scientific outputs and validation acceptance thresholds were not modified.

The result now connects clean training to certified geometry and operational noisy detection in one fixed-load family. It does not finish arbitrary-load or arbitrary-importance theory, establish a real-model result, or certify scientific novelty. Further toy work should resolve a specific prediction or transfer blocker, rather than accumulate more special-case proofs. The two global analytic statements, their computer-assisted examples, and the unfavorable detection outcomes are the evidence to carry forward together.

8.10 The remaining paper-critical decisions

The completed mechanism is clean population reconstruction objective $\rightarrow$ selected geometry $\rightarrow$ fixed decoder $\rightarrow$ noisy detection. The research direction remains the sparsity/load/importance

tradeoff, but the proved transition fixes load at two and importance ratio at two. Independent pair-block replication would merely add their risks; it would not derive a new load-dependent crossing or prove global optimality among unrestricted larger codes.

The motivating monosemanticity paper uses different learning objectives, continuous toy amplitudes, and corruption locations. This binary, fixed-energy, code-noise calculation is a controlled extension of that motivation, not an automatic strict generalization of all its results. A focused comparison of these exact assumptions and claims with direct prior theory remains necessary. Verification does not settle novelty.

The next real-model test must specify the feature measurement, corruption location, decoder/outcome and comparator before substantial compute is allocated. An SAE dictionary Gram matrix is not automatically the tied autoencoder map in this chapter. A transfer experiment may test the qualitative prediction that clean reconstruction selection and downstream corrupted detection need not agree, without claiming that the toy CDF formula is exact in a diffusion network. New root-count proofs, additional noise families, or recursive diffusion retraining are not needed to finish the current bounded mathematical task.

8.11 Does the reconstruction reversal survive decoder calibration?

The clean-optimal representation and its clean-optimal decoder are two different objects. A corrupted input distribution may make a previously optimal decoder bias inappropriate even when its representation is fixed. To distinguish this effect from a representation cost, allow the same decoder adjustment for both shared and monosemantic codes, and compare the same continuous reconstruction loss before and after adjustment.

8.11.1 A fixed, globally solved comparison

Let b_1, b_2 be independent Bernoulli(1/2) variables, with equal feature importance. There is one code dimension and two concepts. Write $w = (w_1, w_2)$, $\|w\|_2^2 = 1$, and $G = w^\top w$. The clean-optimal shared geometry already derived is

$$a_* = \frac{1 + 1/\sqrt{2}}{2}, \quad c_* = \frac{1 - 1/\sqrt{2}}{2}, \quad r_* = \sqrt{a_* c_*}. \quad (8.34)$$

We retain both relative-sign representatives $w_+ = (\sqrt{a_*}, \sqrt{c_*})$ and $w_- = (\sqrt{a_*}, -\sqrt{c_*})$. The monosemantic comparator is $w_{\text{mono}} = (1, 0)$; exchanging the equally important concepts gives the same population outcome. Its omitted concept is estimated by its prior, not omitted from the loss.

Their clean-optimal biases, ordered as strong then weak column, are

$$\beta_{+,1} = \frac{2(c_* - r_*)}{3}, \quad \beta_{+,2} = \frac{a_* - r_*}{2}, \quad (8.35)$$

$$\beta_{-,1} = \frac{2c_* + r_*}{3}, \quad \beta_{-,2} = \frac{a_* + r_*}{2}, \quad (8.36)$$

$$\beta_{\text{mono}} = (0, 1/2). \quad (8.37)$$

The globally selected shared clean loss is $(9 + 2\sqrt{2})/48$, strictly below mono's 1/4. This is a population optimum statement, not a statement that every projected-Adam trajectory reaches it.

Freeze each encoder and add scalar Gaussian code noise:

$$h = wb + \sigma Z, \quad Z \sim \mathcal{N}(0, 1), \quad \hat{b}_i = \max\{w_i h + \beta_i, 0\}. \quad (8.38)$$

The primary outcome throughout is

$$L(w, \beta; \sigma) = \sum_{i=1}^2 \mathbb{E}[(\hat{b}_i - b_i)^2]. \quad (8.39)$$

No binary decision threshold enters this comparison. Noise is in code space, and its projected standard deviation for feature i is $s_i = \sigma|w_i|$.

8.11.2 Gaussian moments and the scalar calibration objective

Condition on a binary state b , put $z_{i,b} = (Gb)_i$, $y_{i,b} = b_i$, and $\mu_{i,b} = z_{i,b} + \beta_i$. For $s > 0$, define the first and second rectified Gaussian moments

$$M_1(\mu, s) = \mathbb{E}[(\mu + sZ)_+] = s\phi(\mu/s) + \mu\Phi(\mu/s), \quad (8.40)$$

$$M_2(\mu, s) = \mathbb{E}[(\mu + sZ)_+^2] = (\mu^2 + s^2)\Phi(\mu/s) + \mu s\phi(\mu/s). \quad (8.41)$$

Here ϕ and Φ are the standard normal density and distribution function. These formulas follow by integrating over $Z > -\mu/s$: $\int_a^\infty z\phi(z) dz = \phi(a)$ and $\int_a^\infty z^2\phi(z) dz = 1 - \Phi(a) + a\phi(a)$. At $s = 0$, use $M_1 = \mu_+$ and $M_2 = \mu_+^2$ directly.

For one feature, its scalar population loss is

$$f_i(\beta) = \sum_b P(b) \left[M_2(z_{i,b} + \beta, s_i) - 2y_{i,b}M_1(z_{i,b} + \beta, s_i) + y_{i,b}^2 \right], \quad P(b) = \frac{1}{4}. \quad (8.42)$$

Since the feature biases are independent optimization coordinates, the oracle-calibrated fixed-geometry loss is

$$L_{\text{cal}}(w; \sigma) = \min_{\beta_1} f_1(\beta_1) + \min_{\beta_2} f_2(\beta_2). \quad (8.43)$$

Oracle population calibration uses the true noisy distribution and is not a finite-data deployment procedure.

8.11.3 Derivatives and a uniform curvature bound

Differentiation of the moments gives $\partial_\mu M_1 = \Phi(\mu/s)$ and $\partial_\mu M_2 = 2M_1$. Thus, for $s_i > 0$,

$$f'_i(\beta) = 2 \sum_b P(b) [(\mu_{i,b} - y_{i,b})\Phi(\mu_{i,b}/s_i) + s_i\phi(\mu_{i,b}/s_i)], \quad (8.44)$$

$$f''_i(\beta) = 2 \sum_b P(b) \left[\Phi(\mu_{i,b}/s_i) - \frac{y_{i,b}}{s_i} \phi(\mu_{i,b}/s_i) \right]. \quad (8.45)$$

The second derivative can be negative, so a single bounded local minimizer does not establish the global optimum. Using $0 \leq y_{i,b} \leq 1$, $\Phi \leq 1$, and $\phi \leq 1/\sqrt{2\pi}$ yields the globally valid bound

$$|f''_i(\beta)| \leq H_i := 2 \left(1 + \frac{1}{s_i\sqrt{2\pi}} \right). \quad (8.46)$$

This deliberately conservative constant applies at every finite bias.

For an interval with midpoint m and radius q , Taylor's theorem and the curvature bound imply

$$f_i(\beta) \geq f_i(m) + f'_i(m)(\beta - m) - \frac{H_i}{2}(\beta - m)^2 \geq f_i(m) - |f'_i(m)|q - \frac{H_i q^2}{2}. \quad (8.47)$$

The second inequality minimizes the concave quadratic lower expression over the interval, whose minimum lies at an endpoint. Nonnegativity of MSE supplies another lower bound, zero.

8.11.4 Explicitly excluding the two infinite tails

Let $U_i = 1 - \min_b z_{i,b}$. For $\beta \geq U_i$ all conditional means satisfy $\mu_{i,b} \geq 1 \geq y_{i,b}$. Every term in $f'_i(\beta)$ is positive when $s_i > 0$. Hence the right half-line has minimum at U_i .

For the left half-line choose, before optimization,

$$B_i = -\max_b z_{i,b} - 12s_i - 1. \quad (8.48)$$

The rectified first moment is nondecreasing in its mean, while the second moment is nonnegative. Since $y_{i,b}^2 = y_{i,b}$ for binary labels, for every $\beta \leq B_i$,

$$f_i(\beta) \geq p_i - 2 \sum_b P(b) y_{i,b} M_1(z_{i,b} + B_i, s_i), \quad p_i = \sum_b P(b) y_{i,b} = \frac{1}{2}. \quad (8.49)$$

The selected finite boundary is useful only if this lower bound exceeds an actually attainable loss. That comparison is checked and saved; it is not assumed from the size of the multiplier 12. If either exclusion fails, the prescribed control remains unresolved rather than silently enlarging its search.

For an exactly zero encoder column, $z_{i,b} = s_i = 0$, so its decoder is constant. The exact optimum is $\beta_i = p_i$ with loss $p_i(1 - p_i)$. No noisy scalar solver is needed for that column.

8.11.5 The declared numerical branch-and-bound calculation

A bounded local minimizer, the clean bias, and the two finite boundary values provide feasible incumbent losses. Local minimization is used only to obtain an upper bound. The interval $[B_i, U_i]$ is then partitioned adaptively, always expanding an interval with the smallest current lower bound from Equation (8.47). Each midpoint is a new feasible incumbent candidate; intervals whose lower bounds exceed the best upper bound are discarded. Both excluded-tail bounds remain in the global comparison.

The declared floating-point slack is 10^{-12} : numerical lower bounds are reduced and feasible upper values increased by that amount. Stop when the numerical global upper-minus-lower gap is at most 10^{-8} per feature, or after 100,000 interval expansions. An unresolved gap or excluded-tail check is preserved as a failure. Every interval, midpoint loss and gradient, lower bound, pruning action, remaining interval and boundary comparison is archived.

What the numerical bound does and does not mean. The floating slack is not outward-rounded interval arithmetic and does not make this an exact-rational noisy-optimum theorem. It is a numerical global-gap diagnostic, accompanied by independent gradient and Gaussian-quadrature checks. The sign of a model comparison must be separated from the sum of its global gaps and integration discrepancies before being interpreted as evidence.

8.11.6 Why both clean-optimal relative signs have the same profile

For the positive-sign code, a feature-1 score is $a_*b_1 + r_*b_2 + \beta_1$. Replace the other fair bit by $b'_2 = 1 - b_2$; the score becomes $a_*b_1 - r_*b'_2 + (\beta_1 + r_*)$, with target b_1 unchanged. Likewise replacing b_1 by $1 - b_1$ in the feature-2 score converts $r_*b_1 + c_*b_2 + \beta_2$ into $-r_*b'_1 + c_*b_2 + (\beta_2 + r_*)$, with target b_2 unchanged. These transformations preserve the marginal conditional population, and Gaussian projected noise has the same variance. Consequently, feature by feature,

$$f_{i,-}(\beta + r_*) = f_{i,+}(\beta). \quad (8.50)$$

The clean and globally calibrated summed losses therefore agree for both signs. The two transformations need not be applied as one joint state transformation: the outcome is a sum of marginal feature losses. Exchanging equally important features similarly preserves the total.

8.11.7 Prespecified outcome

Only $\sigma = 0, .30, .60$ were evaluated in this control. The zero-noise row is the clean analytical anchor; the two positive levels were already present in the frozen-decoder record. The following values are summed population reconstruction MSE:

Code and decoder policy	0	.30	.60
Sharing, frozen clean biases	.246425565099	.318682818682	.512993808968
Mono, frozen clean biases	.250000000000	.317463880602	.505449771089
Sharing, oracle calibrated	.246425565099	.317930513671	.491134733704
Mono, oracle calibrated	.250000000000	.312505410963	.467092806999
Calibrated sharing minus mono	-.003574434901	+.005425102707	+.024041926705

Both sign variants of sharing agree to floating precision. All ten nonzero-column scalar calibrations meet the per-feature gap target; each required 47–58 interval expansions. The two zero-column calibrations are exact. Summed numerical upper-minus-lower gaps are:

Code	$\sigma = .30$	$\sigma = .60$
Same-sign sharing	$1.142385357 \cdot 10^{-8}$	$1.354560977 \cdot 10^{-8}$
Opposite-sign sharing	$1.142385345 \cdot 10^{-8}$	$1.354560977 \cdot 10^{-8}$
Mono	$6.987186149 \cdot 10^{-9}$	$9.931263173 \cdot 10^{-9}$

The sum gap for two features may exceed 10^{-8} because the stopping target was declared per feature. Twenty analytical-gradient versus central-difference checks have maximum discrepancy $9.1433 \cdot 10^{-11}$. All excluded-region checks pass. The independently integrated risk and tail verification is recorded separately from these student global-gap results.

The independent checker imports neither the Gaussian-moment implementation nor the calibration solver. It integrates each conditional squared loss directly over $Z \in [-12, 12]$, splitting at zero and any rectifier cutoff inside that interval, and bounds the omitted tails by the inequality derived earlier in this chapter. All 85 archived hashes match. Its 36 per-feature frozen/calibrated loss checks differ from the saved values by at most 1.943×10^{-16} ; the largest omitted-tail bound is 3.913×10^{-31} . It also checks the curvature constants, interval lower bounds, excluded-region inequalities and final active-interval global bounds. These checks independently corroborate the calculation; they do not convert the floating-point optimization bounds into an exact interval proof.

The resulting numerical comparison brackets are

$$.0054250912846 \leq L_{\text{cal}}(w_+; .30) - L_{\text{cal}}(w_{\text{mono}}; .30) \leq .0054251096957, \quad (8.51)$$

$$.0240419131602 \leq L_{\text{cal}}(w_+; .60) - L_{\text{cal}}(w_{\text{mono}}; .60) \leq .0240419366371. \quad (8.52)$$

The numerical sign margins are much larger than the optimization gaps and integration discrepancies.

Oracle calibration improves both representations, but mono improves more. The same-outcome noisy disadvantage of sharing therefore survives this control rather than disappearing. This supports a cost within the fixed geometry and tied-ReLU decoder class; it does not prove robustness against every possible readout.

Scope of the result. No encoder was retrained for corruption, no arbitrary-load law was established, and no real-model transfer is represented by this oracle population exercise. The monosemantic code estimates its dropped concept by its prior, with that coverage loss included in the MSE. The result is neither a novelty certificate nor a conference acceptance prediction. Its next use is a reviewed mechanism claim and, only under a separate protocol, one focused real-model test.

8.11.8 A linear reference separates generic overlap from rectification

Removing the rectifier changes the decoder class. In that reference class allow a mean-correcting intercept $d = (I - G)\mathbb{E}[b]$ and predict $\hat{b} = Gb + w^\top \sigma Z + d$. Since $\|w\|^2 = 1$, $G^2 = G$ and $\text{tr } G = 1$; also $\text{Cov}(b) = I/4$. Independence and zero-mean noise eliminate the cross term, giving

$$L_{\text{lin}} = \text{tr}\{(I - G) \text{Cov}(b)(I - G)^\top\} + \sigma^2 \|w\|^2 \quad (8.53)$$

$$= \frac{1}{4} \text{tr}(I - G)^2 + \sigma^2 = \frac{1}{4} + \sigma^2. \quad (8.54)$$

Thus all energy-one rank-one geometries tie in this linear dense reference. Rectification is necessary for the difference in the specified nonlinear comparison. This is analytical context, not another experimental arm and not proof that rectification uniquely explains every contribution to the nonlinear difference. An arbitrary freely learned decoder is outside the present comparison.

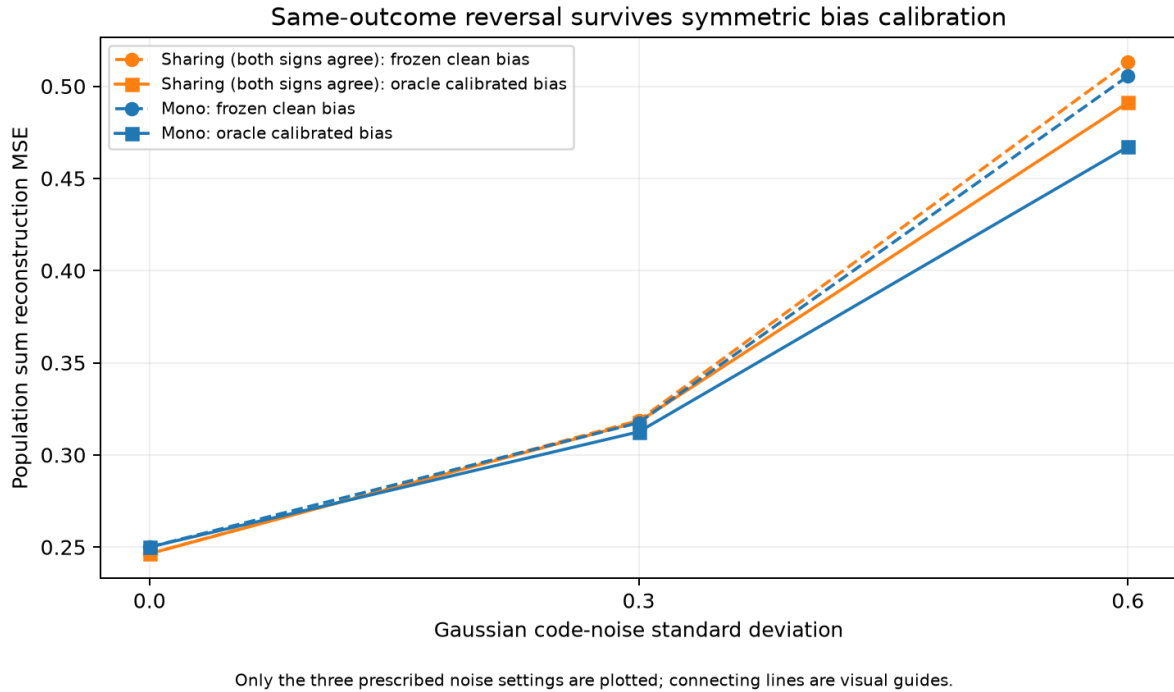


Figure 8.1: Summed population reconstruction MSE at the three prescribed code-noise settings. Sharing has a clean advantage but a corrupted disadvantage under both frozen and oracle-calibrated biases. Both sharing signs give the same population loss. Lines connect saved points for readability; their intersections do not estimate a crossing location or establish a complete robustness boundary.

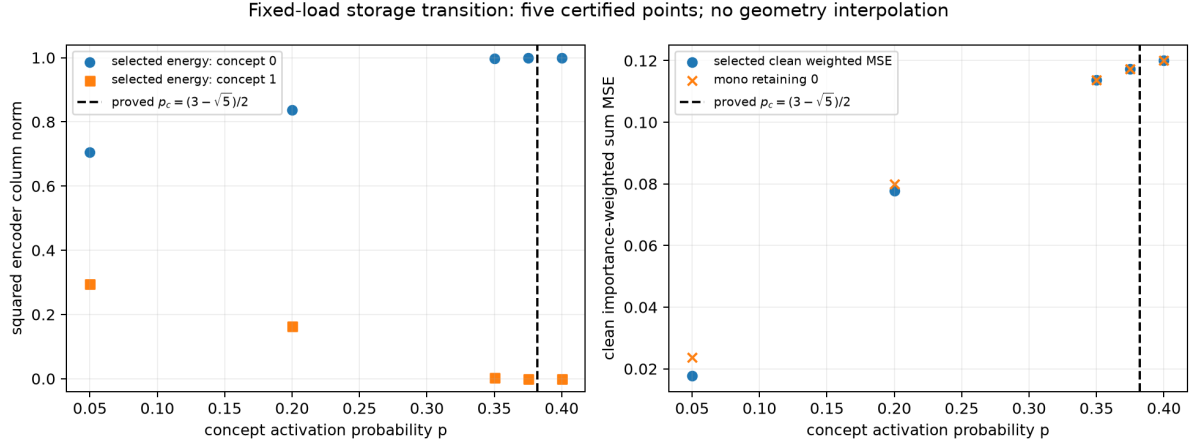


Figure 8.2: Certified clean-selected geometries at five frequencies. Left: encoder energy allocated to the weaker concept. Right: selected and mono clean weighted reconstruction loss. The dashed line is the analytically proved clean-storage transition. The discrete markers do not supply a continuous formula for the selected sharing geometry.

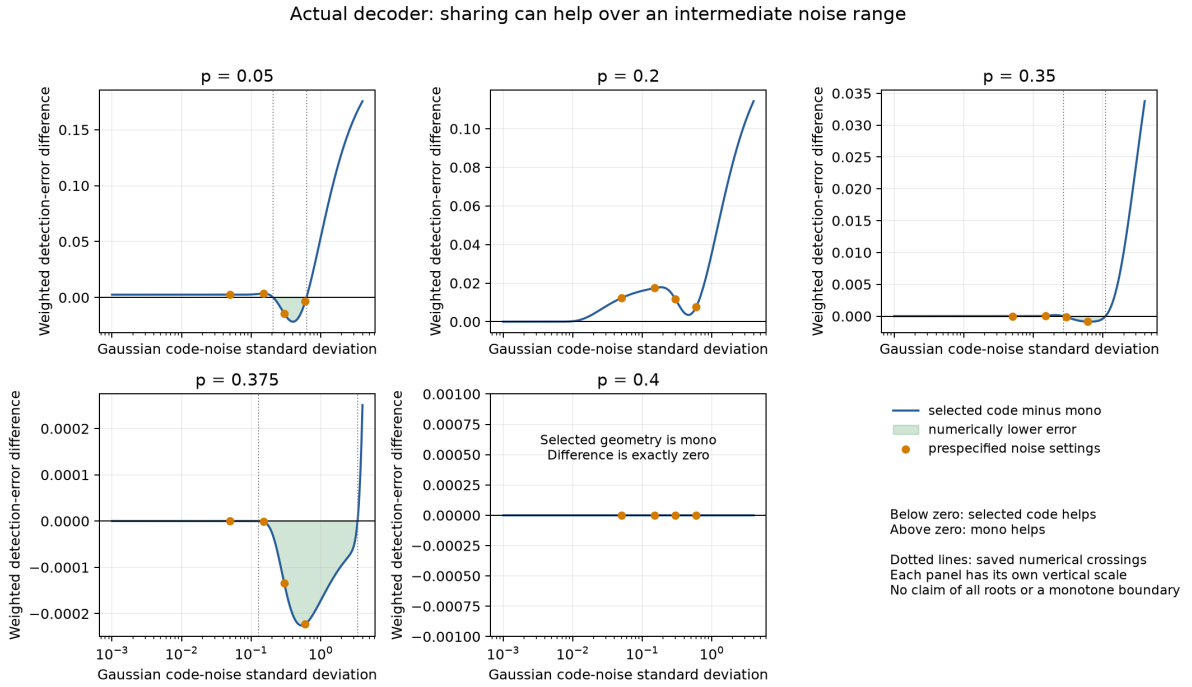


Figure 8.3: Actual-decoder selected-minus-mono weighted detection error versus Gaussian code-noise standard deviation. Negative values favor sharing. Panels use different vertical scales. Dotted lines are saved numerical crossings; they do not certify exact locations or complete counts. The .40 panel is identically zero because both compared codes are mono.

Chapter 9

Project 2: the correct meaning of the covariance recursion

Learning objective. Prove the fixed-map identity, identify its spectral regimes, and avoid mistaking it for a theorem about retraining.

9.1 A fixed linear map

Suppose a population is repeatedly transformed by $x_{g+1} = Gx_g$, with the same symmetric G at every generation. If $C_g = \text{Cov}(x_g)$, then

$$C_{g+1} = GC_gG, \quad C_g = G^g C_0 G^g. \quad (9.1)$$

The proof is simply $\text{Cov}(Ax) = A \text{Cov}(x) A^\top$, followed by induction. Means evolve separately as $\mu_g = G^g \mu_0$.

Diagonalize $G = U \Lambda U^\top$ and let $\tilde{C}_0 = U^\top C_0 U$. Then

$$(U^\top C_g U)_{kl} = (\lambda_k \lambda_l)^g (\tilde{C}_0)_{kl}. \quad (9.2)$$

A direction with initially positive variance has immediate loss if $\lambda_k = 0$, gradual decay if $0 < \lambda_k < 1$, preservation if $\lambda_k = 1$, and growth if $\lambda_k > 1$. Positive semidefiniteness does not imply $\lambda_k \leq 1$.

If C_0 is positive definite, total variance tends to zero exactly when all eigenvalues are below one. If C_0 is singular, only directions on its support matter; a growing eigenvalue with no initial variance does not contribute. Even this elementary statement needs the support qualification.

9.2 Why rank deficiency does not imply gradual decay

A rank- m projector has eigenvalues one and zero. Applying it once discards its nullspace; applying it again changes nothing. Thus a bottleneck can cause a one-time loss without progressive generational collapse. A compression argument alone cannot establish a rate of repeated degradation.

Moreover, eigenvectors are generally mixtures of named concepts. A rare concept indexed by i does not automatically have an eigenvalue “ λ_i ” that determines its fate. Track its projections onto the eigenspaces, or directly track its decoded probability.

9.3 The same margin can mean shrinkage, stability or growth

Consider

$$G_c = \frac{c}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad c > 0. \quad (9.3)$$

Both features have $M_i = 1/\sqrt{2}$ for every c . The eigenvalues are c and zero. With $C_0 = I$, for $g \geq 1$ the trace is c^{2g} . For $c = 1/2$ it decays, for $c = 1$ it is constant after the first step, and for $c = 2$ it explodes.

Therefore the same normalized leakage score cannot determine recursive stability. The two questions can share a matrix without being controlled by the same scalar. Counting eigenvalues exactly equal to one is also not a robust monosemanticity measure: a tiny global rescaling changes the count while leaving the directions and M_i unchanged.

9.4 Adaptive maps and injected variation

If $x_{g+1} = A_g x_g + \epsilon_g$, with independent zero-mean ϵ_g of covariance Q_g , then

$$C_{g+1} = A_g C_g A_g^\top + Q_g. \quad (9.4)$$

Writing $P_{g:k} = A_{g-1} \cdots A_k$ and $P_{k:k} = I$ gives

$$C_g = P_{g:0} C_0 P_{g:0}^\top + \sum_{k=0}^{g-1} P_{g:k+1} Q_k P_{g:k+1}^\top. \quad (9.5)$$

Changing maps need not commute, so one cannot raise a single initial Gram matrix to a power. For nonnormal general maps, per-step eigenvalues alone do not control transient amplification; products and singular values matter.

9.5 Finite-sample shrinkage is estimator-specific

For N independent samples from a distribution with covariance C , the maximum-likelihood-style covariance estimator with denominator N and an estimated sample mean satisfies

$$\mathbb{E}[\hat{C} \mid C] = \frac{N-1}{N} C. \quad (9.6)$$

This follows from subtracting the random sample mean. The unbiased estimator with denominator $N-1$ has expectation C . These facts do not mean that every generative training loop has a universal shrinkage factor $1 - c/N$.

If a particular simplified recursion is $\mathbb{E}[C_{g+1} \mid C_g] = \kappa G C_g G$, then its expected solution is $\mathbb{E}C_g = \kappa^g G^g C_0 G^g$ for fixed G . Setting $G = I$ gives $\kappa^g C_0$, not $\kappa^{2g} C_0$. Almost-sure behavior and expected behavior may differ; neither should be substituted for the other.

Status of this result. The covariance identities are valid and useful baselines. They describe the specified linear processes. They do not prove that diffusion self-training obeys a fixed Gram-matrix map, or that monosemanticity must decay.

Project 2: an exact nonlinear generation operator

Learning objective. Use the same feature decoder from Project 1 to define and solve a genuine repeated-generation toy process.

10.1 Define the transition before discussing collapse

For the detector of Chapter 4, let

$$K_g(b' | b) = \mathbb{P}_\xi\{\text{decode}(W_g b + \xi) = b'\}. \quad (10.1)$$

This is a Markov kernel: every column is nonnegative and sums to one. If P_g is the distribution of current features, the synthetic distribution is

$$Q_{g+1} = K_g P_g. \quad (10.2)$$

With real-data replacement fraction $\alpha \in [0, 1]$, define

$$P_{g+1} = \alpha P_0 + (1 - \alpha) K_g P_g. \quad (10.3)$$

This is a completely specified nonlinear-decoder self-consumption model. At fixed $W_g = W$, it repeatedly passes data through the same channel. It becomes a retraining model only when a learning rule for W_g is supplied, as in Chapter 12.

10.2 The pair kernel in closed form

Use the code $h = a(b_1 - b_2) + \xi$ from Chapter 5 and state order 00, 01, 10, 11. For a source state b , set $d = a(b_1 - b_2)$. Then

$$K(01 | b) = \Phi\left(\frac{-a/2 - d}{\sigma}\right), \quad (10.4)$$

$$K(10 | b) = \Phi\left(\frac{d - a/2}{\sigma}\right), \quad (10.5)$$

$$K(00 | b) = 1 - K(01 | b) - K(10 | b), \quad (10.6)$$

$$K(11 | b) = 0. \quad (10.7)$$

The last equality is architectural: the two threshold events cannot occur together. Project 1's one-generation error and Project 2's redistribution are now computed from exactly the same conditional events.

At zero noise, the states 00, 01, 10 reproduce themselves, while 11 maps to 00. If the initial inputs are independent, the joint probability $p_1 p_2$ is lost immediately. Subsequent iterations cause no additional change. This is a clear feature-combination loss, but it is not progressive decay of all features and does not require changing G .

10.3 An explicit order parameter for a defined loss

For any pair (i, j) with nonzero initial co-occurrence, define joint retention

$$r_{ij,g} = \frac{\mathbb{P}_g(b_i = b_j = 1)}{\mathbb{P}_0(b_i = b_j = 1)}. \quad (10.8)$$

For the antipodal pair with $\alpha = 0$, $r_{12,g} = 0$ for all $g \geq 1$. With mixing, $\mathbb{P}_{g+1}(11) = \alpha \mathbb{P}_0(11)$, so $r_{12,g} = \alpha$ for $g \geq 1$. This is an exact indicator of a particular lost structure, not a universal definition of monosemanticity.

Marginal retention $\mathbb{P}_g(b_i = 1)/\mathbb{P}_0(b_i = 1)$ measures something else. Noise can increase a marginal through false positives while destroying correct combinations. A useful collapse evaluation needs both marginal and joint fidelity, plus semantic correctness.

10.4 What the eigenvalues mean here

At fixed K , $P_g = K^g P_0$ without mixing. A column-stochastic matrix always has eigenvalue one, representing a stationary distribution. Eigenvalues inside the unit disk govern how distributions approach stationary behavior under suitable irreducibility and aperiodicity conditions; reducible chains can have several stationary classes.

Stationarity is not the same as fidelity to P_0 . A process can converge stably to the wrong distribution. Thus a “stability eigenvalue” alone does not measure useful concept survival. Track the stationary bias and the transient rate separately.

For $\alpha > 0$, the affine fixed point is

$$P_* = \alpha \{I - (1 - \alpha)K\}^{-1} P_0 = \alpha \sum_{k=0}^{\infty} (1 - \alpha)^k K^k P_0. \quad (10.9)$$

The inverse exists because $(1 - \alpha)K$ has spectral radius below one. This mixture over channel depths makes the effect of real-data replenishment transparent.

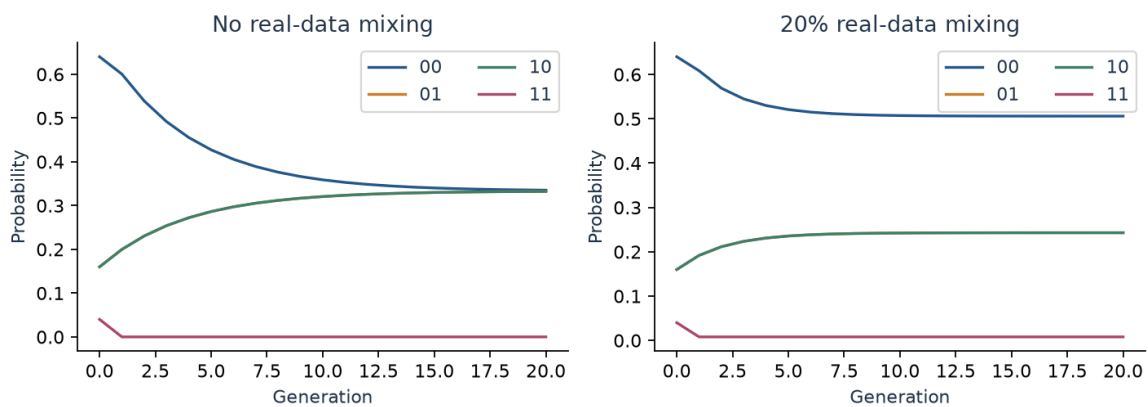


Figure 10.1: Exact pair-kernel calculations with $a = 1$, $\sigma = 0.35$ and initial activation probability 0.2. Marginal frequencies can rise even though the joint state 11 is eliminated from generated outputs. Twenty-percent real-data mixing preserves some joint mass through replenishment.

10.5 A monosemantic comparator

A code retaining only feature 1 with zero output for feature 2 also has a kernel, but every output with $b'_2 = 1$ has zero probability. It drops all of feature 2, not only its simultaneous occurrence with feature 1. This is the generative analogue of the Project 1 capacity tradeoff: a more isolated code can be accurate on its retained concept while representing less of the full distribution.

The comparison should therefore ask how much meaningful information survives at fixed capacity, rather than declaring any noninterfering representation automatically superior.

Before moving on. Compute the zero-noise pair distribution after one and two generations. Explain why covariance alone would not be the clearest account of the lost joint concept.

Chapter 11

The mathematical link between the two projects

Learning objective. Prove how one-generation feature errors bound accumulated distribution drift, and identify what the bound cannot say.

11.1 Use the generation mechanism as a coupling

Under the kernel of Chapter 10, sample $b \sim P_g$ and then $b' \sim K_g(\cdot \mid b)$. This constructs both the current and generated distributions on one probability space. Let

$$e_{i,g} = \mathbb{P}(b'_i \neq b_i), \quad R_g = \sum_i I_i e_{i,g}. \quad (11.1)$$

These are exactly the one-step error quantities analyzed in Project 1, evaluated under the current distribution and geometry.

Theorem 11.1 (Feature-error control of distribution drift). *For any such binary-feature generation kernel,*

$$|\mathbb{P}(b'_i = 1) - \mathbb{P}(b_i = 1)| \leq e_{i,g}, \quad (11.2)$$

$$\text{TV}(K_g P_g, P_g) \leq \mathbb{P}(b' \neq b) \leq \sum_i e_{i,g}. \quad (11.3)$$

Consequently, without real-data mixing,

$$\text{TV}(P_t, P_0) \leq \min \left(1, \sum_{g=0}^{t-1} \sum_i e_{i,g} \right). \quad (11.4)$$

Proof. For a binary coordinate, $|\mathbb{E}(b'_i - b_i)| \leq \mathbb{E}|b'_i - b_i| = e_{i,g}$. For any event A , the indicators of $b \in A$ and $b' \in A$ can differ only if $b \neq b'$, so the difference in their probabilities is at most $\mathbb{P}(b' \neq b)$. Taking the supremum over A gives the coupling bound for total variation. A union bound over coordinates gives the next inequality. Finally use the triangle inequality across generations, and the fact that total variation is at most one. $\square$

11.2 Importance-weighted version

Define the weighted Hamming cost $d_I(b, b') = \sum_i I_i |b_i - b'_i|$. The optimal-transport distance using this cost is no larger than the cost of our particular coupling:

$$W_{d_I}(P_g, K_g P_g) \leq R_g. \quad (11.5)$$

If all $I_i \geq I_{\min} > 0$, then $\sum_i e_{i,g} \leq R_g / I_{\min}$ also converts the weighted risk into a total-variation bound. This may be loose when importance weights span many orders of magnitude. Reporting per-feature and joint errors remains more informative than a single weighted average.

11.3 A replenishment bound

Suppose $\text{TV}(K_g P_g, P_g) \leq \varepsilon_g$, and use Equation (10.3). Write $D_g = \text{TV}(P_g, P_0)$. Convexity and the triangle inequality give

$$D_{g+1} \leq (1 - \alpha)(D_g + \varepsilon_g). \quad (11.6)$$

Starting at $D_0 = 0$ and assuming $\varepsilon_g \leq \varepsilon$ yields, for $\alpha > 0$,

$$D_t \leq \min \left(1, \frac{1 - \alpha}{\alpha} \{1 - (1 - \alpha)^t\} \varepsilon \right). \quad (11.7)$$

This derivation allows K_g to depend on the current distribution; it does not require the same kernel every round. It is a bound on drift under a controlled per-round error, not proof that a small real-data fraction fixes any possible learning algorithm.

11.4 When the bound is informative

If errors are rare and concentrated in a small set of features, the result links a measurable one-step mechanism to a finite-horizon prediction. It can also be vacuous: summing many errors or many generations can exceed one. A stronger paper would exploit transition structure, cancellation, feature dependencies or rare-state hazards to obtain sharper predictions than the union bound.

The implication runs one way. A small coupling error guarantees small drift. A large error does not force drift: a process can swap states while preserving their overall distribution. Conversely, nearly unchanged marginal probabilities can hide lost co-occurrences. These are testable limitations that help choose order parameters.

11.5 Where monosemanticity enters the actual research claim

Geometry influences the one-step error through the exact risk or interference bound. The generation kernel propagates those errors. The candidate scientific chain is therefore

$$\text{feature organization} \longrightarrow \text{specific decoding errors} \longrightarrow \text{recursive distribution change}. \quad (11.8)$$

The theorem establishes the middle-to-last link for the specified toy mechanism. The first link requires either a geometry-controlled theorem or an intervention. The real-model experiment must test the corresponding chain rather than merely correlate an SAE statistic with generation number.

This is a mathematically coherent way to combine the projects. The coupling inequalities themselves are standard probability tools; the publishable contribution would need to be the nontrivial geometry-dependent prediction, its sharpness and its empirical support.

Before moving on. Construct a two-state process that flips both states with equal probability mass. Its per-sample error is one but its distribution does not change. Explain why this does not contradict the theorem.

What changes when the model is genuinely retrained

Learning objective. Write a learning operator and derive its local response, rather than assuming that geometry stays fixed.

12.1 Training is part of the dynamics

Let θ collect encoder, decoder and threshold parameters. For a distribution P on toy states, define a regularized population objective

$$\mathcal{L}(\theta; P) = \sum_b P(b) \ell(\theta, b) + \lambda \Omega(\theta), \quad \theta^*(P) \in \arg \min_{\theta} \mathcal{L}(\theta; P). \quad (12.1)$$

An explicit tie-breaking or optimization rule is needed if there are several solutions. A reconstruction or smooth detection surrogate can be used for training; if it differs from the evaluation risk, state that difference.

The population training-and-generation map is

$$F(P) = \alpha P_0 + (1 - \alpha) K(\theta^*(P)) P. \quad (12.2)$$

This includes refitting the model each generation. A finite-sample version also samples a dataset, estimates parameters and may warm-start from the previous generation. Those choices affect the map and should be logged.

12.2 Deriving the response of the learned parameters

Assume a differentiable local optimum satisfying $\nabla_{\theta} \mathcal{L}(\theta^*; P) = 0$, with nonsingular Hessian $H = \nabla_{\theta}^2 \mathcal{L}(\theta^*; P)$. For a distribution perturbation δP with $\sum_b \delta P(b) = 0$, differentiate the stationarity condition:

$$H \delta \theta + \sum_b \delta P(b) \nabla_{\theta} \ell(\theta^*, b) = 0. \quad (12.3)$$

Thus

$$D_P \theta^*[\delta P] = -H^{-1} \sum_b \delta P(b) \nabla_{\theta} \ell(\theta^*, b). \quad (12.4)$$

This is an implicit-function calculation. It is local and assumes a smooth, identifiable branch of solutions. Symmetries, singular Hessians, ReLU boundaries and optimizer changes can invalidate its use unless handled by gauge fixing, regularization, smooth surrogates or a more general analysis.

12.3 The local generation Jacobian

Differentiate Equation (12.2):

$$DF(P)[\delta P] = (1 - \alpha) [K\delta P + \{D_{\theta}K[D_P\theta^*[\delta P]]\}P]. \quad (12.5)$$

The first term propagates a changed input distribution through the current decoder. The second term captures how retraining changes the decoder itself. A fixed- G or fixed- K argument omits this second mechanism.

At a smooth fixed point, eigenvalues of this Jacobian on the probability-simplex tangent space provide a local stability criterion: spectral radius below one gives local asymptotic stability under the usual finite-dimensional differentiability conditions. A stable fixed point can still be far from P_0 , so quantify its fidelity separately.

12.4 A tractable staged experiment

Start with a small finite family of candidate geometries, such as isolated versus paired codes with stated norm constraints. Evaluate the training objective exactly under P_g , select the minimizer by a deterministic rule, and generate using its exact kernel. This yields a genuinely adaptive piecewise-defined recursion that can be enumerated and checked.

Then move to learned continuous dictionaries. At each generation, retrain on sampled data and compare observed one-step errors with the prediction conditional on the measured G_g . If the geometry changes, report it. The goal is not to force a fixed-matrix prediction onto a changing model.

12.5 Order parameters worth testing

Use a small set of quantities with distinct meanings: per-feature false negatives and false positives; reference-distribution feature retention; joint-state retention; normalized row leakage; decoder reconstruction fidelity; and the local transition or learning Jacobian when it is estimable. Candidate predictors should be evaluated on later generations or held-out regimes, not selected after seeing all collapse outcomes.

A basis-invariant covariance rank measures occupied linear directions. It does not by itself identify named concepts. A neuron purity measure depends on a semantic reference. Neither should be relabeled as the other. The research opportunity is to show which geometric and semantic measurements add predictive information beyond frequency, norm, ordinary reconstruction error and training loss.

Status of this result. Equations (12.4)–(12.5) are conditional local derivations. Solving a global nonlinear learning recursion and proving that a particular semantic scalar forecasts real-model collapse remain research work.

Diffusion: the bridge that is valid, and the one to test

Learning objective. Understand the denoising equations and why a feature Gram matrix is not automatically a diffusion-generation operator.

Source map: DDPM, score-based generative modeling and the recursive-diffusion references in Appendix D. The Gaussian calculation below is derived independently..

13.1 A diffusion model in five equations

A forward diffusion process gradually adds noise to a clean example x_0 . Let $\beta_\tau \in (0, 1)$, $\alpha_\tau = 1 - \beta_\tau$ and $\bar{\alpha}_\tau = \prod_{k=1}^\tau \alpha_k$. Then

$$q(x_\tau | x_{\tau-1}) = \mathcal{N}(\sqrt{\alpha_\tau}x_{\tau-1}, \beta_\tau I). \quad (13.1)$$

Because a composition of affine Gaussian channels is Gaussian,

$$x_\tau = \sqrt{\bar{\alpha}_\tau}x_0 + \sqrt{1 - \bar{\alpha}_\tau}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (13.2)$$

A neural network is trained to predict the added noise, often with the simplified objective

$$\mathcal{L}_{\text{noise}} = \mathbb{E}_{x_0, \tau, \epsilon} \|\epsilon - \epsilon_\theta(x_\tau, \tau)\|_2^2. \quad (13.3)$$

The pointwise minimizer of squared error is a conditional expectation. The prediction is therefore related to the score of the noisy data distribution, not a literal reconstruction Gram matrix. One common reverse-process mean parameterization is

$$\mu_\theta(x_\tau, \tau) = \frac{1}{\sqrt{\alpha_\tau}} \left(x_\tau - \frac{\beta_\tau}{\sqrt{1 - \bar{\alpha}_\tau}} \epsilon_\theta(x_\tau, \tau) \right). \quad (13.4)$$

Sampling uses a sequence of reverse transitions, often with additional noise:

$$x_{\tau-1} = \mu_\theta(x_\tau, \tau) + s_\tau z, \quad z \sim \mathcal{N}(0, I), \quad (13.5)$$

with a specified variance or sampler convention. The sampler, schedule, truncation and network all affect the generated distribution.

13.2 Two clocks, two different processes

Diffusion time τ indexes noise removal within one generation of one sample. Training generation g indexes fit-sample-refit cycles across models. A denoising step can reduce uncertainty while a correctly sampled full generative process preserves diversity. Confusing these clocks makes “denoising contracts covariance, so diffusion must collapse” an invalid argument.

Write the full recursive process as

$$\theta_g = \text{Train}(D_g), \quad Q_g = \text{SampleLaw}(\theta_g), \quad D_{g+1} \sim \alpha P_0 + (1 - \alpha)Q_g. \quad (13.6)$$

Here data size, optimization budget, initialization, conditioning and sampling parameters are part of the operator. There is no justified substitution $Q_g = G_g P_g$ merely because an SAE dictionary has a Gram matrix G_g .

13.3 A Gaussian denoiser that can be solved completely

Let the generation- g data distribution be $X \sim \mathcal{N}(0, C_g)$ and observe $Y = X + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Gaussian conditioning gives

$$\mathbb{E}[X | Y] = A_g Y, \quad A_g = C_g(C_g + \sigma^2 I)^{-1}. \quad (13.7)$$

This is the optimal population denoiser refitted to the current covariance. If the next generation's data consist of these posterior means, then

$$C_{g+1} = A_g(C_g + \sigma^2 I)A_g^\top \quad (13.8)$$

$$= C_g(C_g + \sigma^2 I)^{-1}C_g. \quad (13.9)$$

The eigenvectors are unchanged and each positive eigenvalue evolves by

$$c_{g+1} = \frac{c_g^2}{c_g + \sigma^2}. \quad (13.10)$$

It decreases strictly when $\sigma^2 > 0$ and converges to zero. To prove the limit, monotonicity gives $c_g \downarrow L \geq 0$; continuity implies $L = L^2/(L + \sigma^2)$, hence $L = 0$. When $c_g \ll \sigma^2$, the next value is approximately c_g^2/σ^2 , explaining rapid late-stage shrinkage.

This is a genuine refitting recursion in an analytically tractable denoising model. It is not a full DDPM theorem and does not yet distinguish monosemantic from rotated Gaussian representations.

13.4 Posterior sampling changes the answer completely

The conditional covariance is

$$V_g = C_g - C_g(C_g + \sigma^2 I)^{-1}C_g. \quad (13.11)$$

If the next sample is drawn from the exact posterior rather than set equal to its mean, the law of total covariance gives

$$\text{Cov}(X') = \text{Cov}(\mathbb{E}[X | Y]) + \mathbb{E}[\text{Cov}(X | Y)] = C_g. \quad (13.12)$$

Thus the same denoiser accompanied by the correct sampling variation preserves the original distribution. Removing sampling variance creates the collapse in Equation (13.9). This is why a real diffusion sampler must be modeled or held fixed, not discarded from the argument.

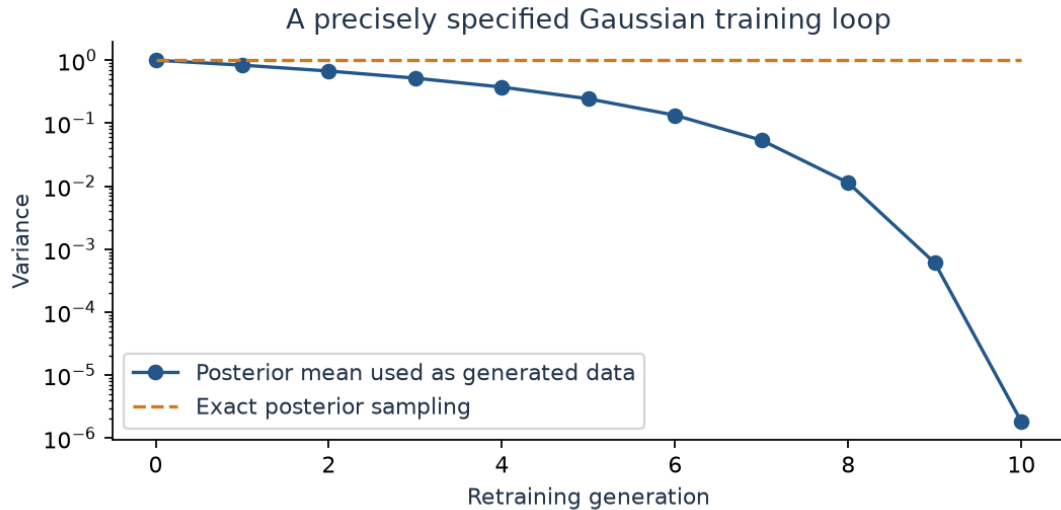


Figure 13.1: Exact Gaussian population refitting with initial variance 1 and observation-noise variance 0.2. Reusing posterior means contracts variance; exact posterior sampling preserves it. This is a didactic denoising model, not a trained image-diffusion experiment.

13.5 Where a representation-level contribution can enter

Gaussian isotropy cannot identify semantic axes: orthogonal rotations preserve the distribution. To study monosemanticity, use known sparse non-Gaussian factors or a controlled image generator with labeled concepts. Then test whether a feature’s interference and one-step error forecast later loss of its frequency or combinations.

On a real model, an SAE decoder dictionary D_g provides $D_g^\top D_g$, a measure of dictionary overlap. It is not the same as a tied encoder-decoder reconstruction map unless those constraints are imposed. A diffusion denoiser’s input-output sensitivity is a Jacobian, generally rectangular, input-dependent and noise-time-dependent. These are different objects and should retain different symbols.

A measured relation between dictionary overlap, causal feature interventions and future degradation would be evidence for the proposed mechanism. Calling the relation exact without measuring the missing decoder and sampler dynamics would not be justified.

13.6 Existing work the new claim must distinguish itself from

The MAD work studies self-consuming generative loops and the role of real-data supply. The accumulating-data work shows that replacement and accumulation can behave differently. A June 2026 paper by Khelifa, Turner and Venkataramanan already gives a spectral characterization of recursive diffusion collapse driven by reverse-process truncation, even under ideal score estimation. Therefore “the first spectral account of diffusion collapse” is not an available claim based on this reading.

The intended distinction is a tested mechanism involving sparse semantic features, interference, robust decoding and subsequent retention. Keep reverse truncation fixed and check whether it explains the apparent effect before attributing that effect to feature geometry.

Before moving on. Identify the covariance term that posterior sampling restores in the Gaussian example. Explain why posterior means alone lose it.

Experiments that test the derivations

Learning objective. Design experiments that can support or falsify the mechanism, with a realistic route from exact toys to a vision diffusion model.

14.1 Stage A: exact identities and finite-state calculations

The accompanying script already checks conditional moments from the anchor paper, pair risks by Monte Carlo, the pair transition matrix, the fixed-linear covariance identity, a same- M counterexample, the Gaussian recursion and the linear tight-frame comparison. These are unit-level mathematical validations, not the final research experiments.

Reproduce the exact pair boundary for several importance ratios and for both scale conventions. For each setting, compare the normal-CDF formula with independent random trials and binomial uncertainty. Include rare-but-important and common-but-low-importance combinations rather than varying all frequencies together.

For the transition model, enumerate the distribution rather than assuming it remains factorized. Track the entire four-state law, joint retention and marginal errors. Check the coupling inequalities directly. This stage should expose mistakes before any expensive training is run.

14.2 Stage B: learn the geometry in controlled toys

Train tied nonlinear autoencoders on planted features with a weighted reconstruction objective. Use a deliberately limited grid: a few loads (for example 1, 1.5, 2 and 4), several activation rates, two or three importance-decay schedules and multiple seeds. The values are starting design choices, not a mandatory exhaustive Cartesian product.

At each checkpoint, measure G , column norms, active-feature coverage, reconstruction error, statistical feature-detection error and an optional separately labeled adversarial metric. Predict risk using the measured geometry before inspecting held-out outcomes. This tests the conditional theory. Separately ask whether geometry itself is predictable from the training controls; that tests the stronger phase-diagram objective.

Necessary comparators include capacity-matched isolated features, a random dictionary, a learned unconstrained dictionary and a semantic or sparsity-promoting intervention. Control encoder norm or report matched-norm sensitivity analyses. Include a clean-performance comparison so a method is not credited merely for dropping difficult concepts.

14.3 Stage C: recursive training on the same toys

For each chosen starting model, generate a fixed-size synthetic dataset and train the next model under a fixed budget. Compare pure replacement with a specified real-data fraction. Keep a

real-only repeated-training control to estimate ordinary training variability. Distinguish warm starts from fresh initialization; choose one primary protocol and treat the other as an ablation.

The central prediction is prospective: use geometry and one-step errors at generation g to predict feature or joint losses at $g + 1$ and later. A model that only explains already-observed collapse is a description, not an onset predictor. Hold out seeds or regions of the sparsity/load grid when assessing prediction quality.

Analyze false positives and false negatives separately. For a rare feature, a small false-positive rate can dominate its marginal frequency. A seemingly preserved activation rate does not establish preserved semantic content.

14.4 Stage D: one controlled image generator

Before a general image dataset, use images generated from known factors: shape identity, location, color and a rare co-occurring attribute, with explicit train and evaluation distributions. This provides real pixel inputs and genuine convolutional or diffusion learning while retaining concept ground truth. It is a mechanistic testbed, not a benchmark contribution.

Train a small image diffusion model, then repeat the same controlled self-training protocol. Use the known rendering factors or a validated held-out evaluator to measure marginal and joint fidelity. Test feature robustness on fixed real reference images under controlled image corruptions or denoising conditions. Avoid defining a concept detector solely from the same model whose collapse is being measured.

14.5 Stage E: a small natural-image diffusion demonstration

Use one tractable vision setting, such as CIFAR-scale images, with a small established diffusion implementation. A pilot should measure wall-clock cost for one generation before committing to a long recursion. The primary comparison might use two representation/training conditions, two real-data fractions and three seeds over a short, informative horizon. The exact number of generations is determined by the pilot and the observed transition, not by a promise made before measuring compute.

This remains a real-model validation arm. If it fails to show the predicted mechanism, report that limitation and investigate the mismatch. It should not be reclassified as optional success evidence only after seeing unfavorable results.

14.6 How to measure geometry consistently

Fix the layer, diffusion-time sampling distribution and reference inputs. Fit a common diagnostic dictionary across checkpoints, or explicitly align independently learned dictionaries and report alignment uncertainty. Otherwise feature permutations, rescalings, splitting and merging can masquerade as semantic decay.

For a common dictionary, report reconstruction fidelity at every checkpoint: a fixed dictionary can fail because the basis changes, even if useful computation is preserved. With checkpoint-specific dictionaries, distinguish actual concept changes from changed coordinate systems. Use direct labeled-factor behavior as a reference wherever possible.

14.7 Interventions that support a mechanism

An intervention should change the hypothesized interference while monitoring clean quality, norms and coverage. Examples include a controlled coherence penalty in a toy model, a sparsity-promoting module, or a targeted direction manipulation with a random-direction control. A penalty that orthogonalizes everything may simply reduce stored information, so compare at matched capacity and inspect feature loss.

The strongest evidence is a linked change: the intervention alters measured geometry, changes the predicted one-step errors and changes later feature retention in the predicted direction. A correlation between low purity and late generations alone is much weaker.

14.8 Evaluation and uncertainty

Report training seeds as independent replications. For rare events, provide sample counts and uncertainty; zero observed events is not proof of zero probability. If a concept has true frequency p , a sample of size N contains none with probability $(1 - p)^N$, approximately e^{-Np} . To make this probability at most δ , one needs roughly $N \geq \log(1/\delta)/p$. This calculation belongs in the sampling plan.

For generative quality, combine a distributional image metric with concept-frequency and joint-frequency measures. Aggregate metrics can miss tail loss. Use held-out real reference data for evaluation, and never select thresholds or order parameters on the final test outcomes without reporting that selection.

14.9 A compact evidence table

Claim	Required evidence	A result that would weaken it
Geometry predicts one-step robustness	Analytic risk or bound agrees with held-out toy errors	Norm or feature frequency explains the effect equally well
Sparsity/load creates a tradeoff	Capacity-controlled risk crossing, plus learned-toy comparison	Crossing vanishes under energy matching
Errors predict recursive loss	Prospective prediction of later marginal or joint changes	Predictor adds nothing beyond current reconstruction error
Monosemanticity contributes causally	Controlled intervention changes the proposed chain	Apparent benefit is entirely due to dropping concepts
Mechanism transfers to diffusion	Same directional prediction and intervention in fixed-sampler experiments	Sampler truncation or evaluator drift explains the result

An eight-week program that preserves both projects

15.1 Status of this execution outline

This is the original September planning outline, retained as background rather than a renewed eight-week allowance. The October Project 1 results appear in Chapters 7–8. The current stop decision is to consolidate the established claims, compare their exact assumptions with the direct prior literature, and specify one bounded real-model transfer. Broad grids, additional toy families and recursive diffusion retraining are not automatically authorized by this historical outline. The full variable-load/importance boundary remains open.

Learning objective. Turn the derivations into deliverables without promising that a theorem outline already guarantees a conference paper.

15.2 One integrated paper structure

The intended paper can be organized around a precise question: how does sparse feature interference determine immediate decoding reliability and the retention of semantic structure across recursive training? Project 1 supplies the risk tradeoff; Project 2 supplies the propagation mechanism and predictive test. The shared toy model makes the connection explicit, while a diffusion demonstration tests how much of it survives outside the controlled setting.

The theoretical thread should remain active throughout: define assumptions, derive predictions, diagnose mismatches and refine the claim. Experimental work is equally part of the scientific argument. The document does not assign people to narrow permanent roles or prescribe authorship.

15.3 Weekly deliverables

Assuming a Monday 21 September 2026 start, eight full weeks end on 15 November; a tenth week ends on 29 November. If the actual kickoff differs, shift dates while preserving the ordering and compute gates.

Week	Dates	Concrete deliverable
1	Sep 21–27	Read the anchor paper using Volume I. Re-run the supplied checks. Fix task, norm budget, corruption location and two primary hypotheses. Run a diffusion cost pilot and a focused related-work check in parallel with mathematical work.

Week	Dates	Concrete deliverable
2	Sep 28–Oct 4	Reproduce pair risk and kernel. Train the first planted-feature model. Produce one robustness prediction plot and one recursive-distribution plot. Record how learned geometry differs from the solved pair.
3	Oct 5–11	Expand a small sparsity/load/importance grid. Validate conditional risk predictions on held-out seeds. Derive or tighten the part of the load-dependent theory supported by observed structure. Start the controlled-image diffusion loop.
4	Oct 12–18	Month-one checkpoint: a precise theorem statement with assumptions, verified toy evidence for both questions, and an operational real-model pipeline. Select the most informative regimes and eliminate redundant sweeps.
5	Oct 19–25	Test prospective collapse predictions and a geometry intervention. Run the primary natural-image diffusion conditions if the pilot supports the budget. Write the theory and methods sections using actual completed results.
6	Oct 26–Nov 1	Complete the primary recursive runs and matched controls. Investigate mismatches with the theory. Compare geometry predictors with frequency, norm, reconstruction-error and training-loss baselines.
7	Nov 2–8	Finish required seeds and decisive ablations. Assemble the complete paper narrative, plots and limitations. Reproduce the main results from clean configuration files.
8	Nov 9–15	Month-two checkpoint: integrated draft, checked proofs, final main experiments, uncertainty estimates and a reproducible artifact. Obtain technical review focused on the most vulnerable assumption.
9	Nov 16–22	If available: resolve reviewer-style objections and improve one transfer or intervention experiment. Avoid opening a second modality or unrelated task.
10	Nov 23–29	If available: finalize revisions, supplementary proofs and release materials; rerun only checks affected by changes.

15.4 What to reduce if the compute budget is tight

Reduce the number of architectures, modalities, redundant grid points and secondary metrics first. Keep the core mathematical question, both toy mechanisms, the decisive controls and one real-model test. A smaller, interpretable experiment is preferable to an underpowered collection of unrelated demonstrations.

If natural-image diffusion training is too expensive, a controlled-image diffusion model still tests a genuine nonlinear generative model with known concepts. Report the narrower generality honestly. Do not silently replace the real-model arm with another analytic plot and describe the original empirical objective as completed.

15.5 What would make the paper substantive

The standard identities in this volume are foundations. The eventual contribution needs a nontrivial result, such as a sharp learned-geometry tradeoff, a predictive feature-retention mechanism that survives controls, or a targeted intervention justified by theory and validated in a generative loop. The combined story should demonstrate why the two results illuminate each other.

An eight-week program can plausibly produce a focused submission if compute and participation support the plan. No document or toy calculation establishes main-track acceptance in advance. A spotlight is a distinction among accepted papers, not an easier fallback venue. The practical objective is a sound, novel and well-supported manuscript; venue decisions follow external review. The next step is to choose the primary corruption and resource convention, reproduce the analytic pair in a learned toy, and execute the diffusion pilot. This continues the same two-project program with explicit assumptions.

Appendix A

Formula ledger: retain, qualify or replace

Earlier statement	Correct status and replacement
$G = W^\top W$	Exact for the tied encoder geometry. An arbitrary SAE or diffusion model need not have this as its input-output map.
$M_i = G_{ii} / \ G_{i:}\ _2$	Exact alignment ratio for nonzero columns. Equals a radius only for a specified normalized linear decision problem.
$M_i < 1$ whenever an attacker can move the score $\sum \rho_{ij}^2 \approx pn/m$ inside the unrestricted attack margin	False. An isolated feature with $M_i = 1$ can still be perturbed.
$pn/m \leq \tau^{-2} - 1$ is a proved universal boundary	Not justified. Use state-averaged risk or a statistical interference variance.
Rare concepts have high M	Not established. Use the exact pair-risk equality or a stated sufficient error bound.
Monosemanticity is the limit $s \rightarrow 1$	Not implied. Frequency does not enter the fixed- G ratio directly.
$C_{g+1} = GC_g G$	Not an identity. Sparse activations can favor superposition; semantic organization and frequency are distinct.
Compression forces all non-retained modes to decay gradually	Exact for repeatedly applying a fixed linear map. Not a generic retraining law.
Eigenvalue-one fraction is a monosemanticity order parameter	False. A projector loses its nullspace in one step and is then stable. Eigenvalues above one can also occur.
κC_g recursion produces κ^{2g}	Unsupported. It is highly scale-sensitive and ignores semantic basis and initial support.
Two valid identities plus toy plots guarantee a paper	Algebraically inconsistent. The factor is κ^g for that covariance recursion.
Recursive synthetic training is automatically improvement	No. Novelty, strength of mechanism, empirical evidence and review determine the contribution.
	No. It is a recursive process; improvement is an outcome to measure.

Appendix B

Further derivations and worked exercises

B.1 Gaussian ReLU moments for continuous reconstruction

Let $Z = d + \sigma U$ with $U \sim \mathcal{N}(0, 1)$, and let ϕ, Φ be the standard normal density and CDF. Direct integration gives

$$A(d) = \mathbb{E} \text{ReLU}(Z) = \sigma \phi(d/\sigma) + d\Phi(d/\sigma), \quad (\text{B.1})$$

$$B(d) = \mathbb{E} \text{ReLU}(Z)^2 = (d^2 + \sigma^2)\Phi(d/\sigma) + d\sigma\phi(d/\sigma). \quad (\text{B.2})$$

For a target amplitude x , the conditional squared error is $x^2 - 2xA(d) + B(d)$. In the antipodal pair, use $d = x_1 - x_2$ for the first reconstruction and $-d$ for the second. Integrating this expression over the specified spike-and-uniform features yields an exact low-dimensional integral for the noisy continuous-amplitude risk. Evaluating an analytically specified integral is not symbolic regression.

To derive A , substitute $Z = d + \sigma U$ and integrate over $U > -d/\sigma$. Use $\int_a^\infty u\phi(u) du = \phi(a)$. For B , expand $(d + \sigma U)^2$ and use $\int_a^\infty u^2\phi(u) du = a\phi(a) + 1 - \Phi(a)$.

B.2 Mixture covariance includes a mean term

If real data have mean μ_0 and covariance C_0 , while generated data have mean μ_Q and covariance C_Q , a mixture with real fraction α has covariance

$$C_{\text{mix}} = \alpha C_0 + (1 - \alpha)C_Q + \alpha(1 - \alpha)(\mu_0 - \mu_Q)(\mu_0 - \mu_Q)^\top. \quad (\text{B.3})$$

The last term is between-component variation. Omitting it can make a drifting mean look like an unexplained covariance change. The finite-state kernel automatically accounts for this when moments are computed from the full law.

B.3 Exercise 1: an exact adversarial perturbation

For a correctly classified x and ℓ_2 budget, show that $\delta_* = -yf(x)\beta/\|\beta\|_2^2$ reaches the boundary. Its norm is $yf(x)/\|\beta\|_2$. This proves tightness of Equation (2.2). A slightly larger displacement crosses the boundary; equality only reaches it, and classification at equality depends on the tie convention.

B.4 Exercise 2: rare feature versus common nuisance

Take a fixed two-feature Gram matrix with unit diagonal and overlap ρ . Then $M_1 = 1/\sqrt{1 + \rho^2}$ independently of p_1, p_2 . But the statistical nuisance variance in the independent centered detector is $\rho^2 p_2(1 - p_2)$. Conditional robustness of feature 1 depends on its competitor's activity, not simply its own rarity. Its unconditional contribution to total risk also depends on its own false-positive and false-negative mixture.

B.5 Exercise 3: why a largest overlap and a sum differ

Compare one nuisance with overlap a and four nuisances with overlap $a/2$. Their total squared overlap is equal. Under isotropic unrestricted feature noise, the readout variance can agree. Under sparse nonnegative activations, their distributions of interference differ: one large jump is not the same random variable as a sum of four smaller jumps. The Bernstein bound records both V_i and B_i to reflect this distinction.

B.6 Exercise 4: solve the mixed fixed kernel

Starting from $P_{g+1} = \alpha P_0 + (1-\alpha)K P_g$, repeatedly substitute the previous step to obtain a finite geometric series plus $(1-\alpha)^g K^g P_0$. The final term vanishes for $\alpha > 0$ in the probability-vector norm, leaving the fixed-point series in Chapter 10. This is a statement about repeated use of a fixed channel; refitting generally changes K and prevents that same closed form.

B.7 Exercise 5: covariance does not identify concepts

Take a two-dimensional isotropic Gaussian and rotate it by 45 degrees. The covariance and distribution are unchanged, but coordinate axes are different. A covariance-based criterion alone cannot say which axes correspond to meaningful concepts. Non-Gaussian planted factors or external semantic measurements are necessary for that question.

B.8 Exercise 6: a sharply stated falsification

Suppose an SAE overlap score predicts late-generation image degradation until reconstruction error and feature frequency are included, after which it adds no held-out predictive information. This does not invalidate the exact toy algebra. It weakens the claim that the chosen geometry measure explains real-model collapse beyond simpler measurements. The paper should report and analyze that distinction rather than call the transfer confirmed.

Appendix C

Numerical verification record

All calculations here were performed on CPU with fixed seeds 20260921 for the main script and 20260922 for the additional checks. The scripts use NumPy, SciPy and exact SymPy algebra; figures use Matplotlib. They do not train a large neural model or establish publication novelty.

Check	Result
Anchor-paper conditional moments	1.5 million simulated examples; all conditional means and variances within 0.002 of the exact values.
Corrected label-noise J crossing	$\eta = 0.2427153958$ at $S = 0.2$.
Input-noise J crossing	$\lambda = 0.5479269441$ at $S = 0.2$.
Pair boundary, equal stored amplitude	$\sigma = 0.6583495016$ at $p_1 = p_2 = 0.2$, equal importance.
Pair boundary, equal encoder energy	$\sigma = 0.2799389129$ for the same probabilities and importance.
Equal-energy pair risk at $p = 0.2$, $\sigma = 0.3$	Exact 0.261396; independent simulation 0.261172 with 500,000 examples.
Pair risk at $(p_1, p_2, I_1, I_2, \sigma) = (.2, .2, 1, 1, .3)$	Exact 0.152641; simulation 0.151474 with 500,000 examples.
Pair risk at $(.1, .4, 1, .3, .7)$	Exact 0.252367; simulation 0.252943.
Pair risk at $(.6, .3, 1, 2, 1)$	Exact 0.900331; simulation 0.901558.
Pair transition kernel	Nonnegative entries; every column sums to one to numerical precision. Marginal-drift checks pass.
Fixed-map covariance identity	Iterative covariance updates agree with the matrix-power expression for random PSD initial covariance.
Same- M counterexample	$M = 1/\sqrt{2}$ for shrinking, stable and expanding maps; spectra and traces confirm all three regimes.
Linear tight-frame comparison	Distributed and mono risks agree: approximately 0.3333333333 in the tested setting.
General five-feature risk and bound	Eight random three-dimensional dictionaries: exact state-enumerated risks agree with independent simulations within 0.005; the stated error bound dominates the exact risks.
Relative J degradation identity	Exact symbolic simplification verifies the positive-factor expression in Volume I; no equation fitting is used.
Gaussian mean-only loop	Initial variance 1 reaches approximately 1.81×10^{-6} after ten refits at noise variance 0.2; posterior sampling preserves variance 1.

The main record is in `research_notes/verification_results.json`, generated by `research_notes/verify_math.py`. Additional checks are recorded in `research_notes/additional_verification.json` by `research_notes/verify_additional.py`. A tolerance assertion checks numerical implementation agreement; it is not a substitute for a proof. The proofs explain why the formulas hold, and the checks help catch algebra, coding and interpretation errors.

Appendix D

Primary references and claim boundaries

1. Zhang et al. *Beyond Interpretability: The Gains of Feature Monosemanticity on Model Robustness*. <https://arxiv.org/abs/2410.21331>. The source anchor; taught in Volume I. It does not analyze recursive diffusion collapse.
2. Elhage et al. *Toy Models of Superposition* (2022). <https://www.anthropic.com/research/toy-models-of-superposition>. The controlled sparse-feature representation framework.
3. Scherlis et al. *Polysemanticity and Capacity in Neural Networks* (2022). <https://arxiv.org/abs/2210.01892>. Its per-feature capacity, Equation (3), is exactly the squared geometric alignment used here; capacity allocation has prior analytic theory.
4. Shumailov et al. *The Curse of Recursion: Training on Generated Data Makes Models Forget* (2023 preprint). <https://arxiv.org/abs/2305.17493>. Prior model-collapse analysis; no universal finite-sample factor is assumed here.
5. Alemohammad et al. *Self-Consuming Generative Models Go MAD* (2023). <https://arxiv.org/abs/2307.01850>. Prior generative self-consumption work; real-data supply and generation choices are central protocol variables.
6. Gerstgrasser et al. *Is Model Collapse Inevitable? Breaking the Curse of Recursion by Accumulating Real and Synthetic Data* (2024). <https://arxiv.org/abs/2404.01413>. Distinguishes accumulation from replacement; supports treating the data policy as part of the problem.
7. Ho, Jain and Abbeel. *Denoising Diffusion Probabilistic Models* (2020). <https://arxiv.org/abs/2006.11239>. Source for the standard diffusion training and sampling framework.
8. Song et al. *Score-Based Generative Modeling through Stochastic Differential Equations* (2020). <https://arxiv.org/abs/2011.13456>. Score-based formulation of generative diffusion.
9. Khelifa, Turner and Venkataramanan. *Recursively Trained Diffusion Models: Limiting Collapse Distribution and Spectral Characterization* (11 June 2026). <https://arxiv.org/abs/2606.13796>. Existing spectral diffusion-collapse work; the abstract and stated contribution were checked for scope. A detailed theorem-by-theorem comparison is still required before claiming novelty against it.

The literature check for these notes is targeted, not an exhaustive novelty certification. The proposed contribution must be positioned by its exact assumptions, mechanism, prediction and empirical evidence. None of the phrases “first”, “strictly generalizes the source paper” or “guaranteed conference paper” follows from the calculations delivered here.

The completed work in this volume is the teaching and derivation foundation. The remaining scientific work is to characterize learned geometry, sharpen predictive claims where possible and validate the same mechanism in a real generative model within the agreed research program.